虚拟数字人技术

夏时洪

中国科学院计算技术研究所

2026-03-12

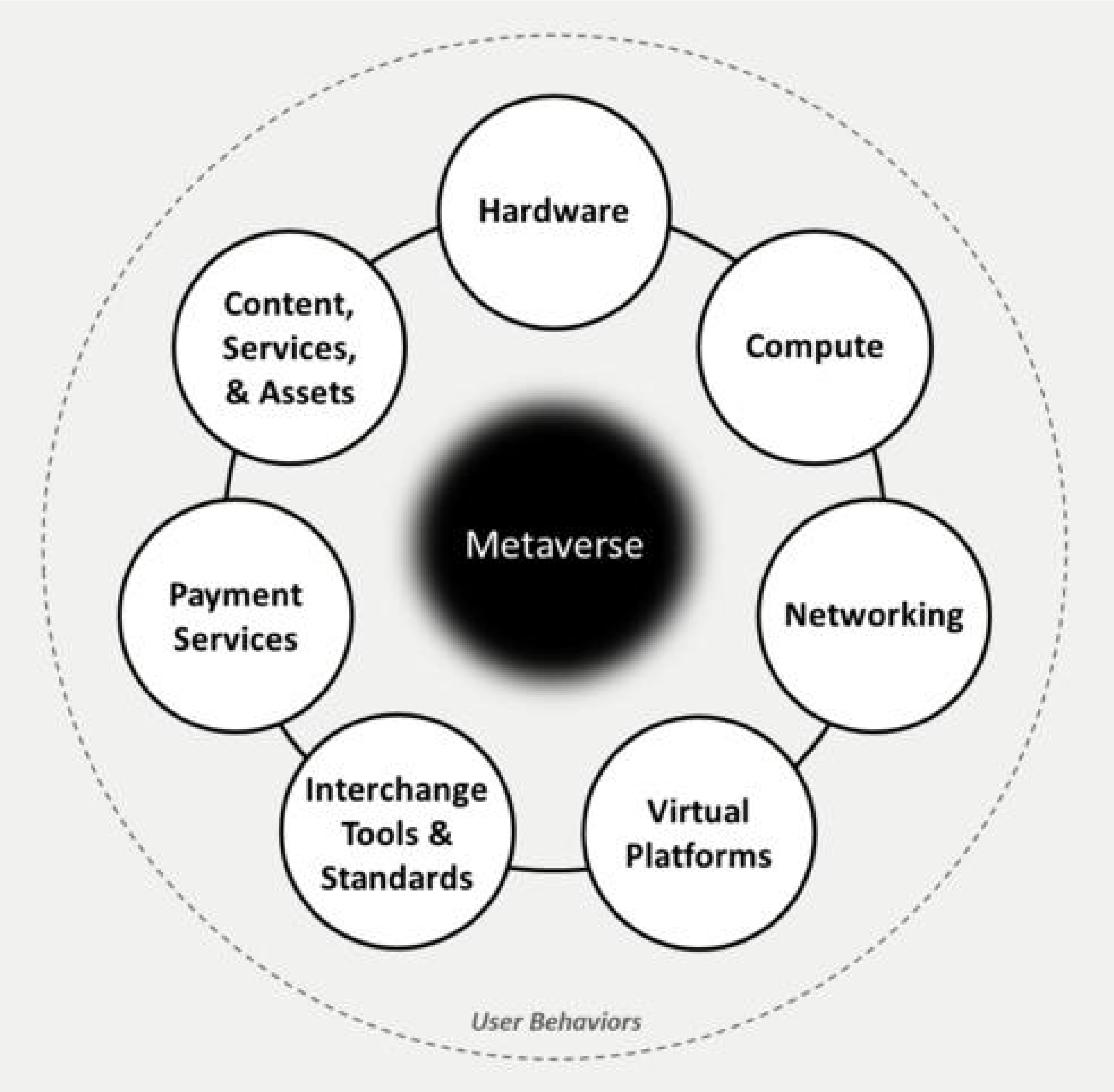
元宇宙1.0

硬件： 用于访问、互动或开发元宇宙的物理技术和设备。

网络：提供持久、实时连接、高带宽和去中心化数据传输。

计算： 提供计算能力以支持元宇宙。如渲染、AI等。

虚拟平台: 开发和运营沉浸式数字内容平台。



交换工具和标准： 工具、协议、格式、服务和引擎。

支付： 对数字支付流程、平台和业务的支持。

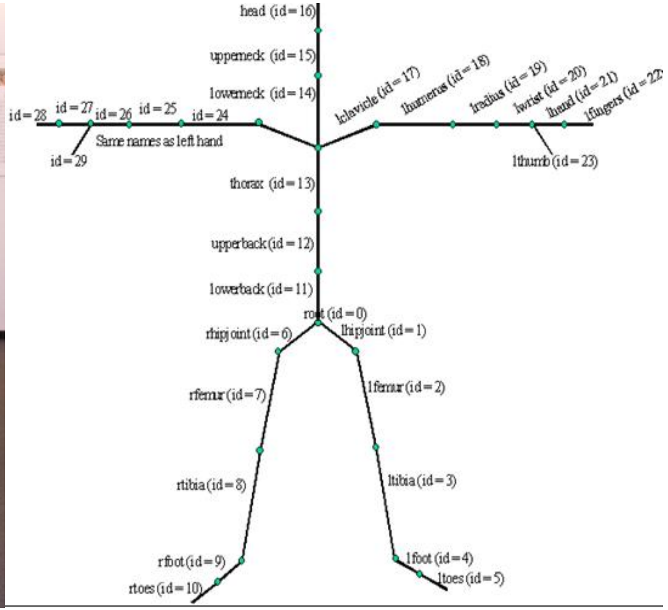
元宇宙内容、服务和资产。

用户行为： 消费者和企业行为（包括消费和投资、时间和注意力、决策和能力）中可观察到的变化。

——By Mathew Ball

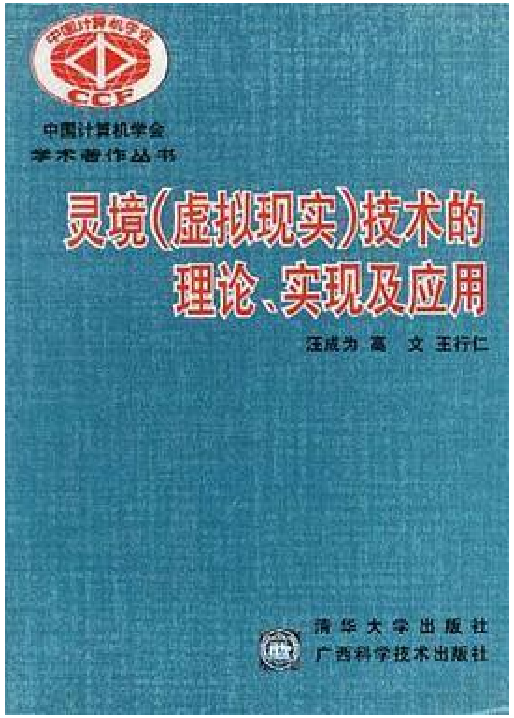
虚拟数字人

1. 2D虚拟数字人，图像视频+DeepFake技术/神经辐射场技术；
2. 3D虚拟数字人，3D骨架/3D人体模型+图形学/动画模拟仿真技术。

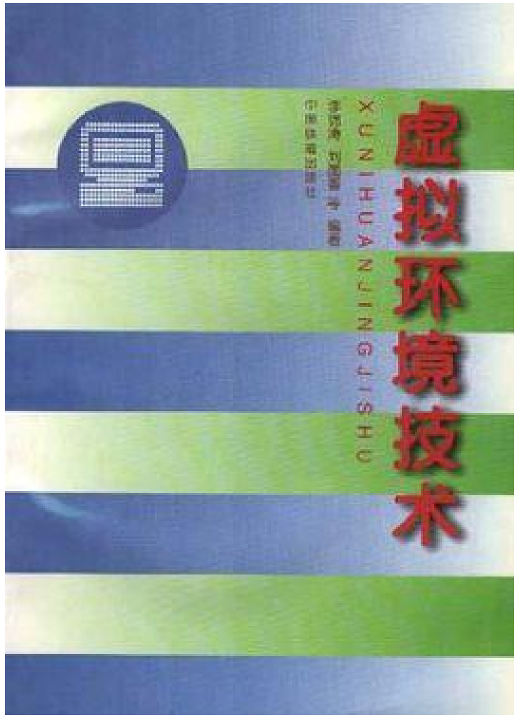





计算所早期代表性工作



汪成为,高文,王行仁(著), 灵境(虚拟现实)技术的理论、实现及应用, 清华大学出版社,1996年



李锦涛等编著，虚拟环境技术，中国铁道出版社，1996年

国家科技进步二等奖2003
基于多功能感知理论的中国手语识别与合成研究
高文、尹宝才、王兆其、陈熙霖、马继涌、王春立、陈益强、宋益波、方高林、杨长水

北京市科学技术奖一等奖2006
面向体育训练的三维人体运动模拟与视频分析系统
王兆其，樊建平，张勇东，夏时洪，李锦涛，李东建，朱登明，毛天露，邱显杰，黄河，唐胜，沈燕飞

报告提纲

1. 代表性科研项目
2. 代表性基础研究

Part 1: 代表性科研项目

代表性工作

体育训练

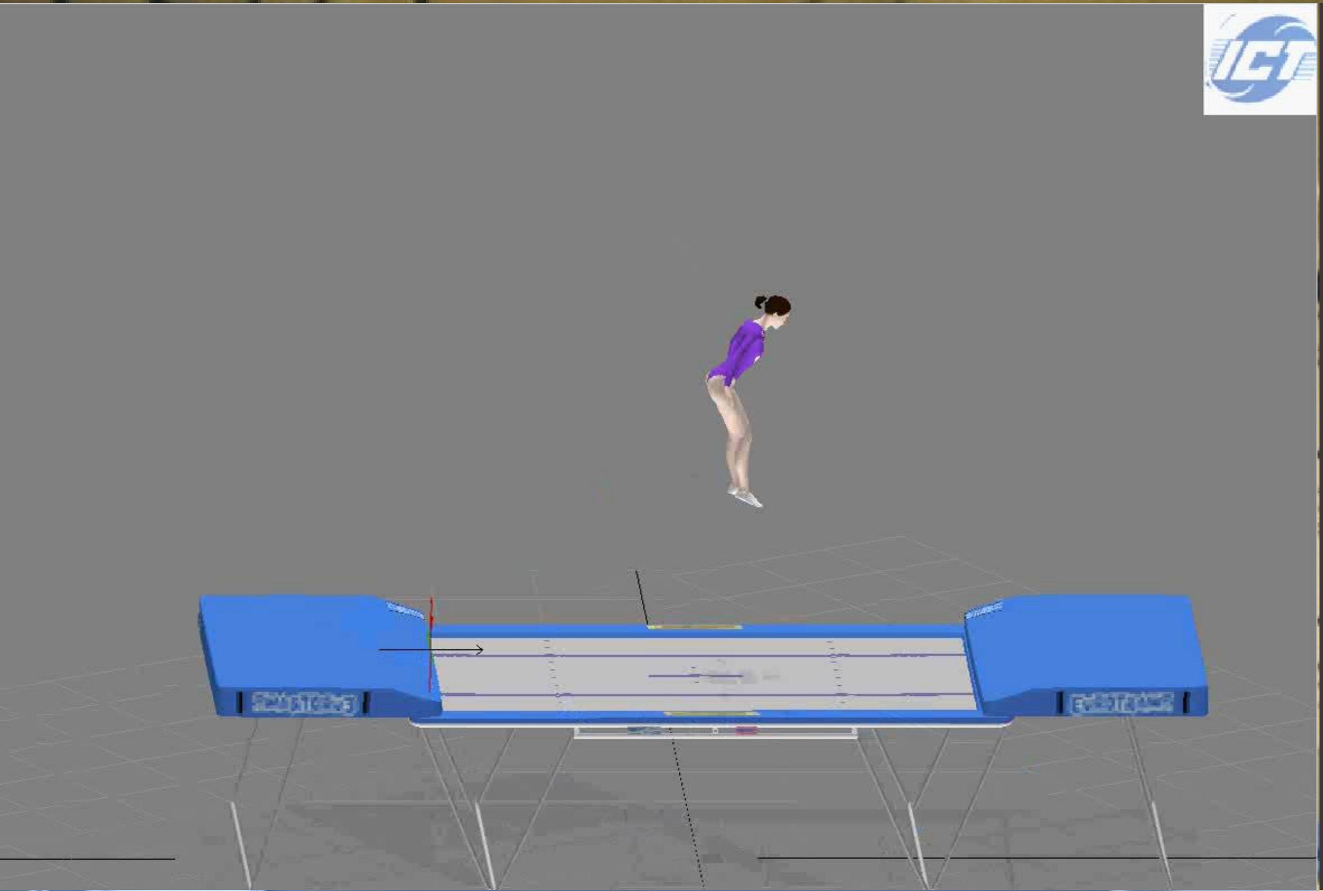
康复训练

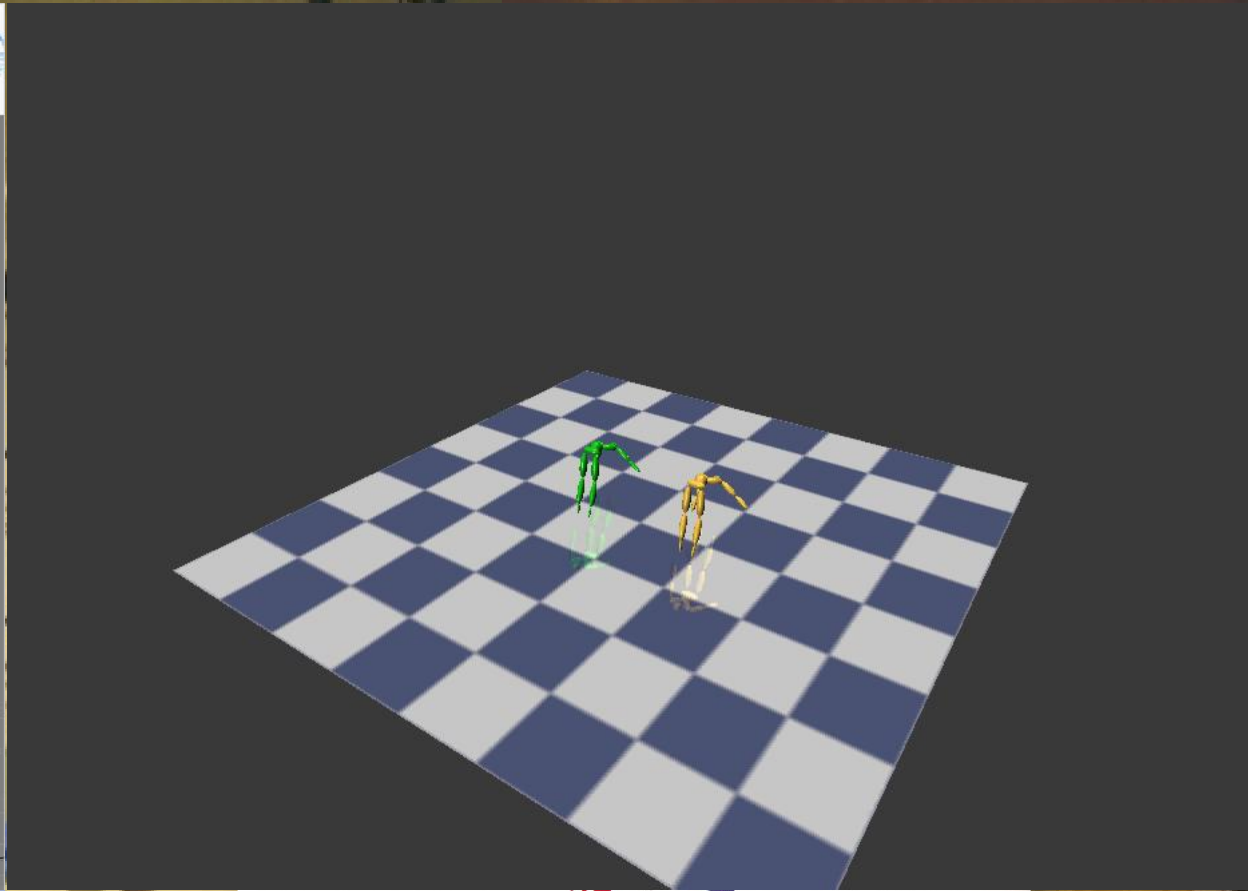
仿真训练

人机工程、界面

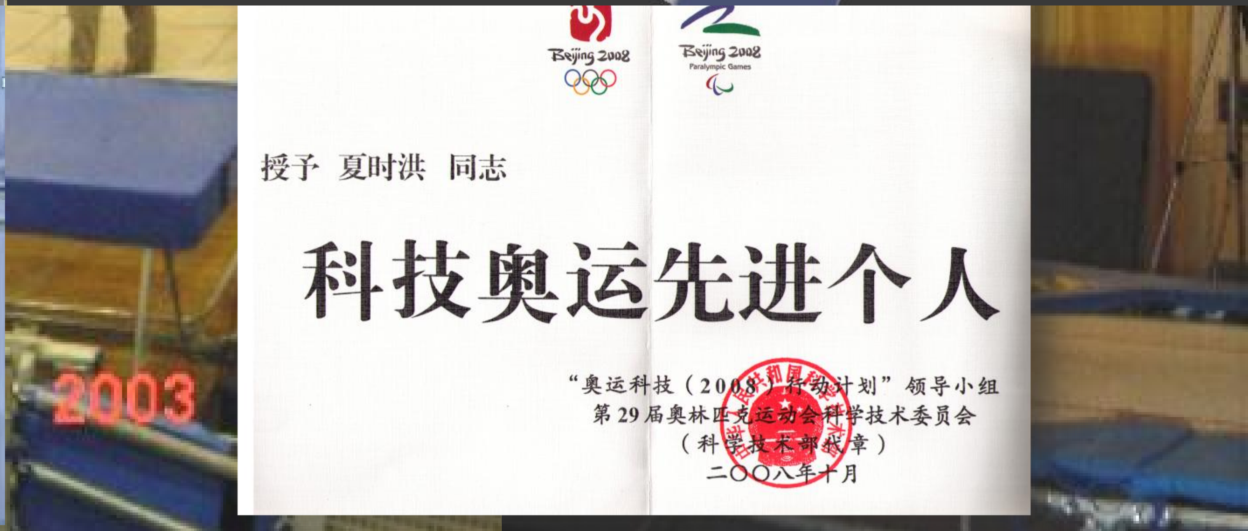

坚持 三从 一
2003 8 15

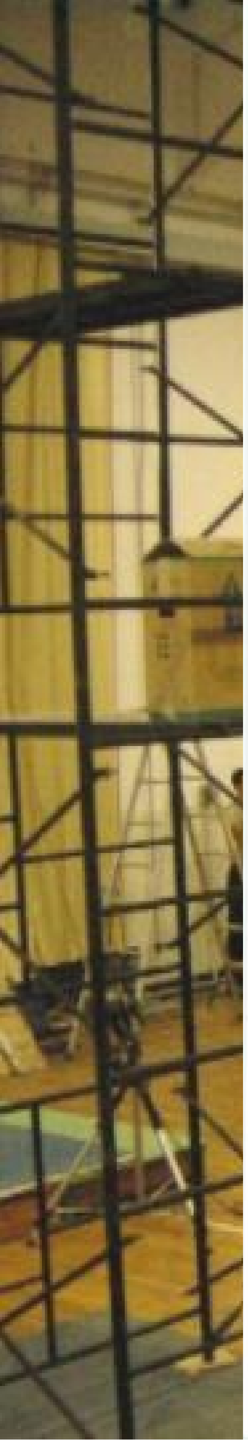
2003 8 16






科院研制成功
维人体运动仿真与分析系统
新闻联播


2003
Beijing 2008
Beijing 2008
授予 夏时洪 同志
科技奥运先进个人
“奥运科技（2008）行动计划”领导小组
第29届奥林匹克运动会科学技术委员会
（科学技术部代章）
二〇〇八年十月



北京市科学技术奖

№ 2006计-1-001-04

为表彰在推动科学技术进步、对首都经济建设和社会发展作出贡献者，特颁此证，以资鼓励。

获奖项目：

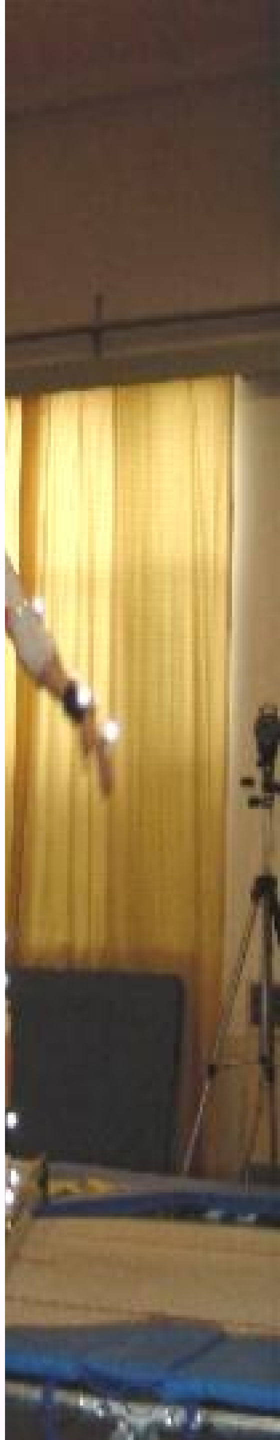
面向体育训练的三维人体运动模拟与视频分析系统

获 奖 者：**夏时洪**

获奖等级：**壹等奖**

北京市人民政府

二〇〇八年十二月




热烈欢迎航天发射任务全系统模拟仿真训练演练观摩活动全体代表


实时


证书

获奖项目：虚拟人技术及其在载人航天中的应用

获奖单位：中国科学院计算技术研究所

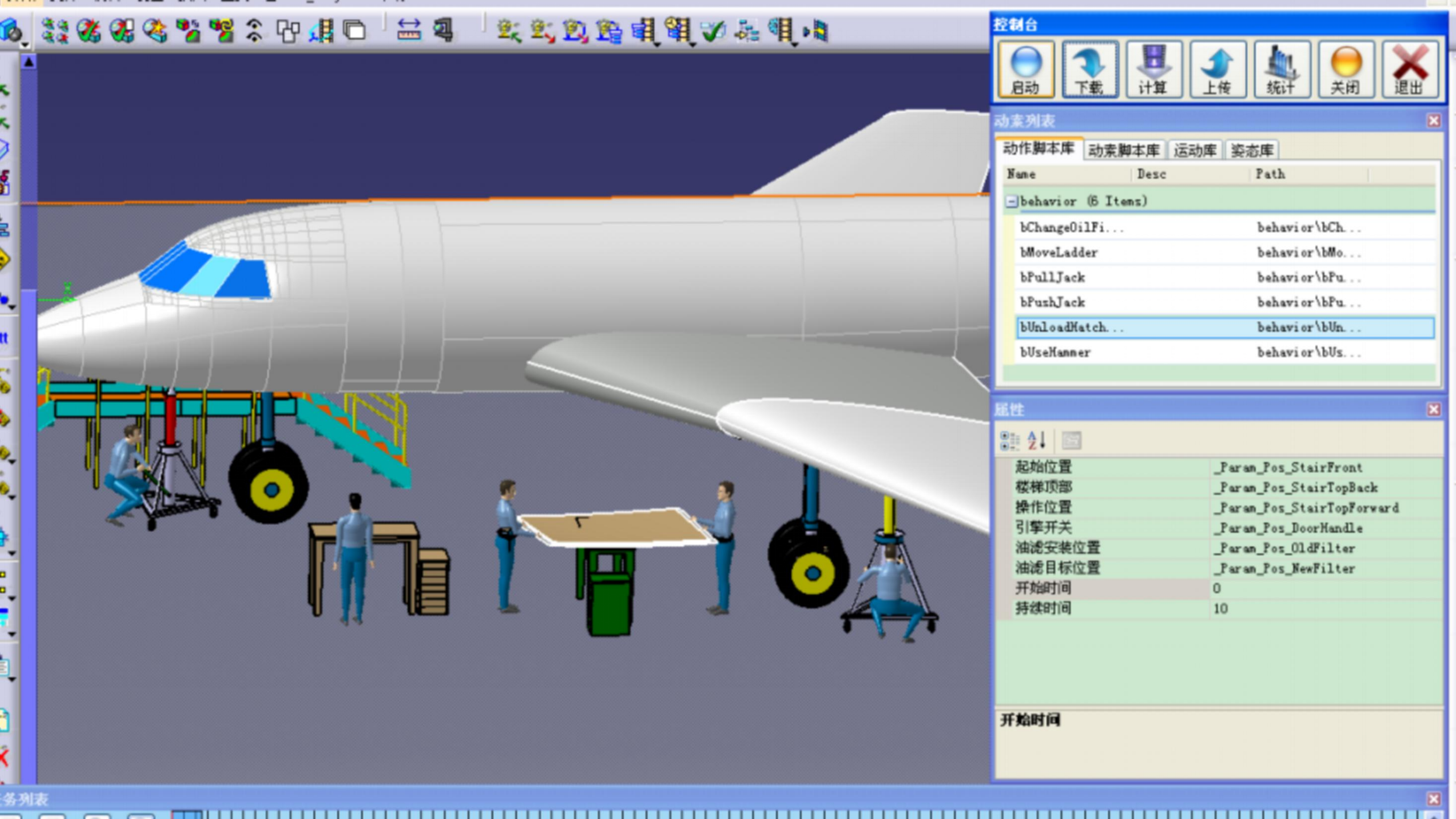
获奖等级：军队科技进步贰等奖

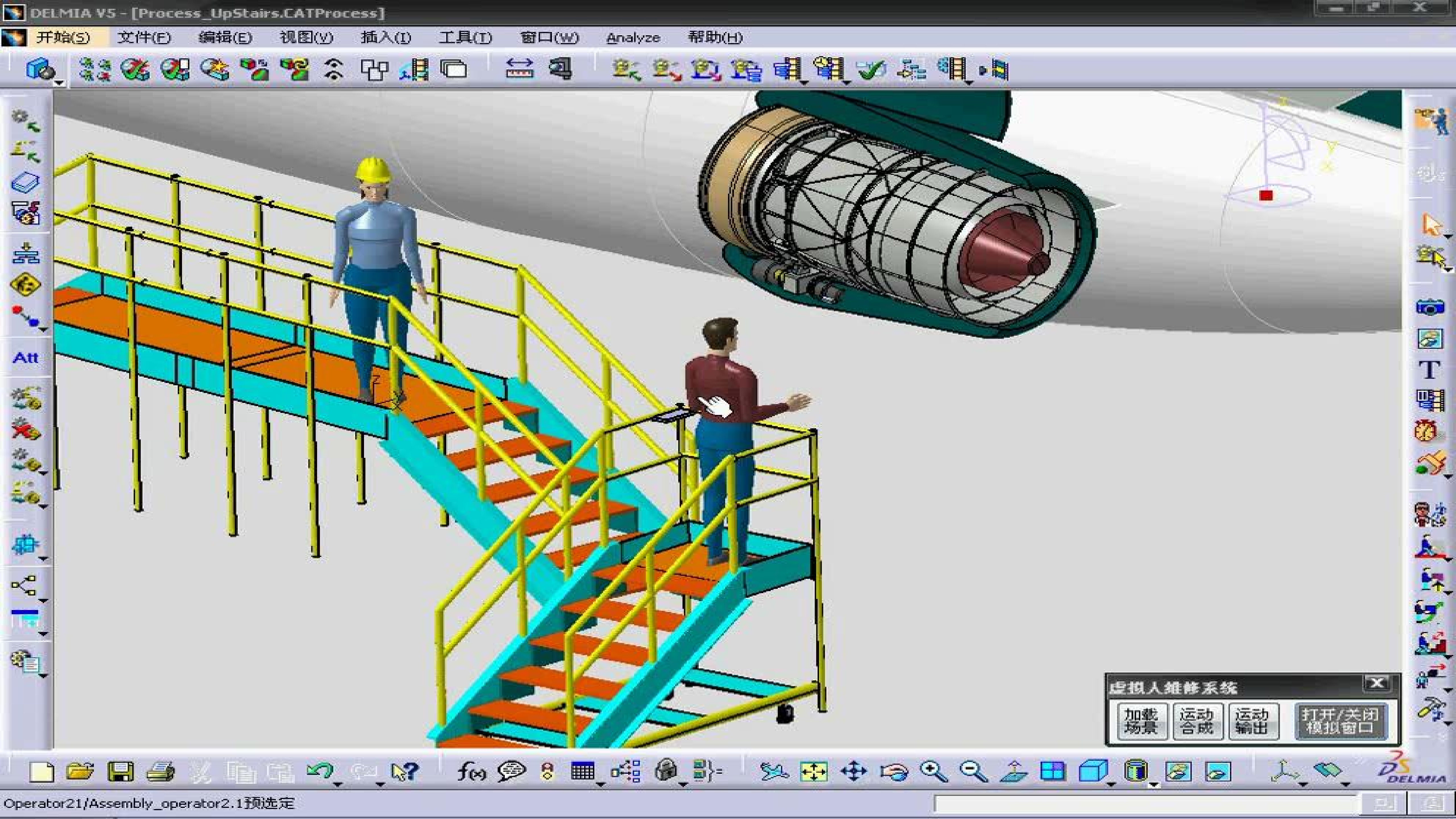
奖励日期：2012 年 11 月

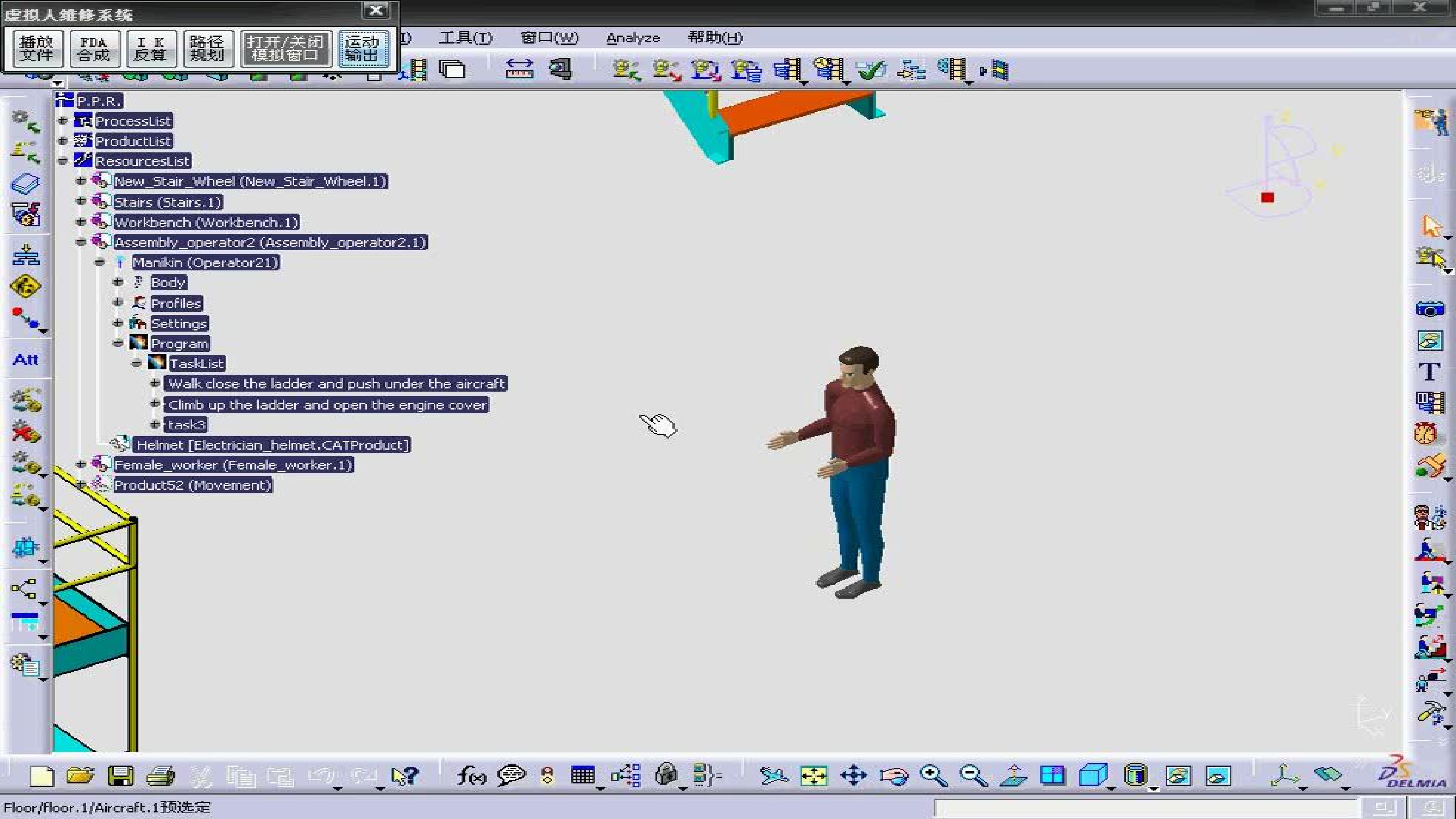
证 书 号：2012921227

总 装 备 部

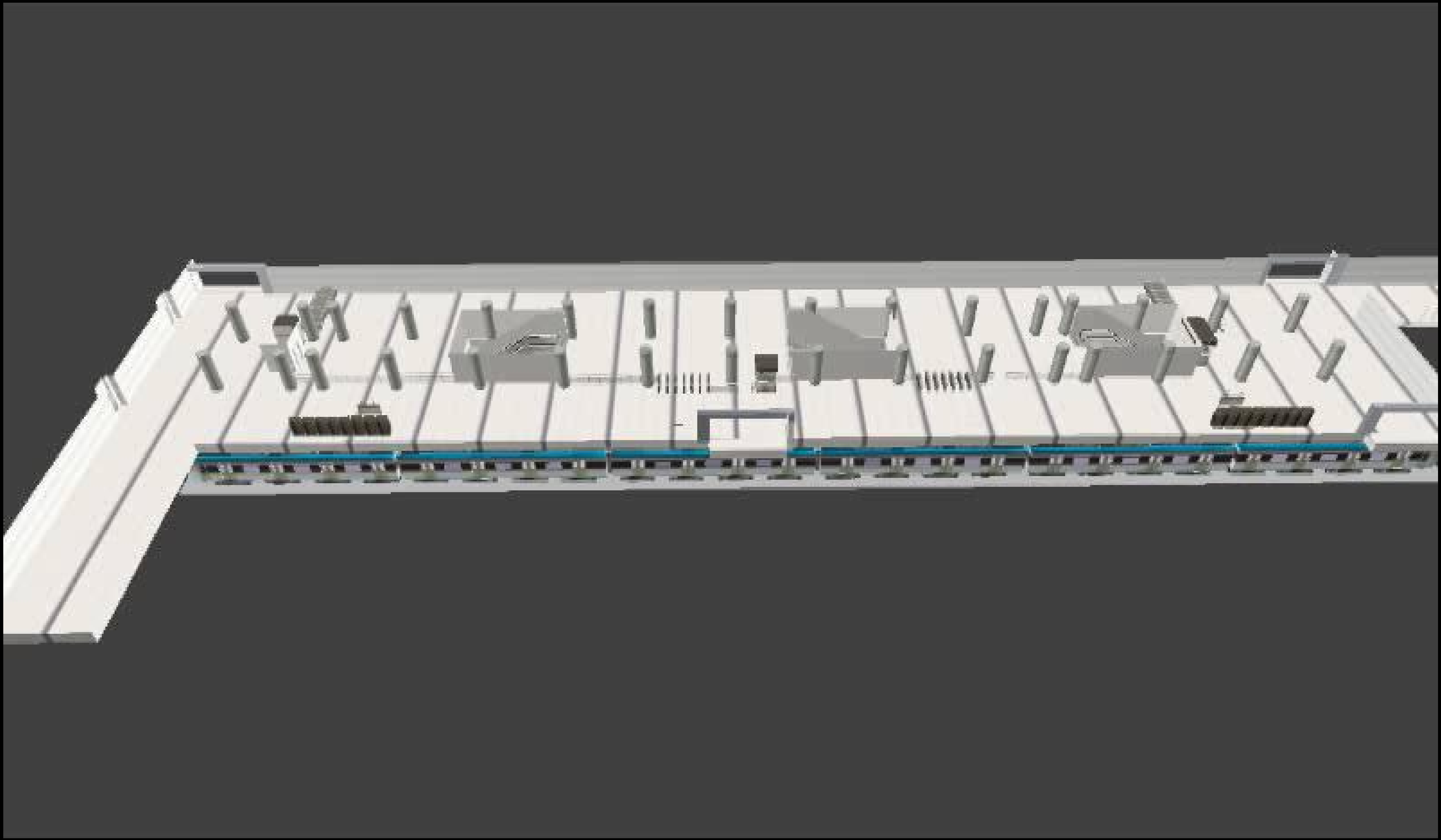
中国人民解放军总装备部

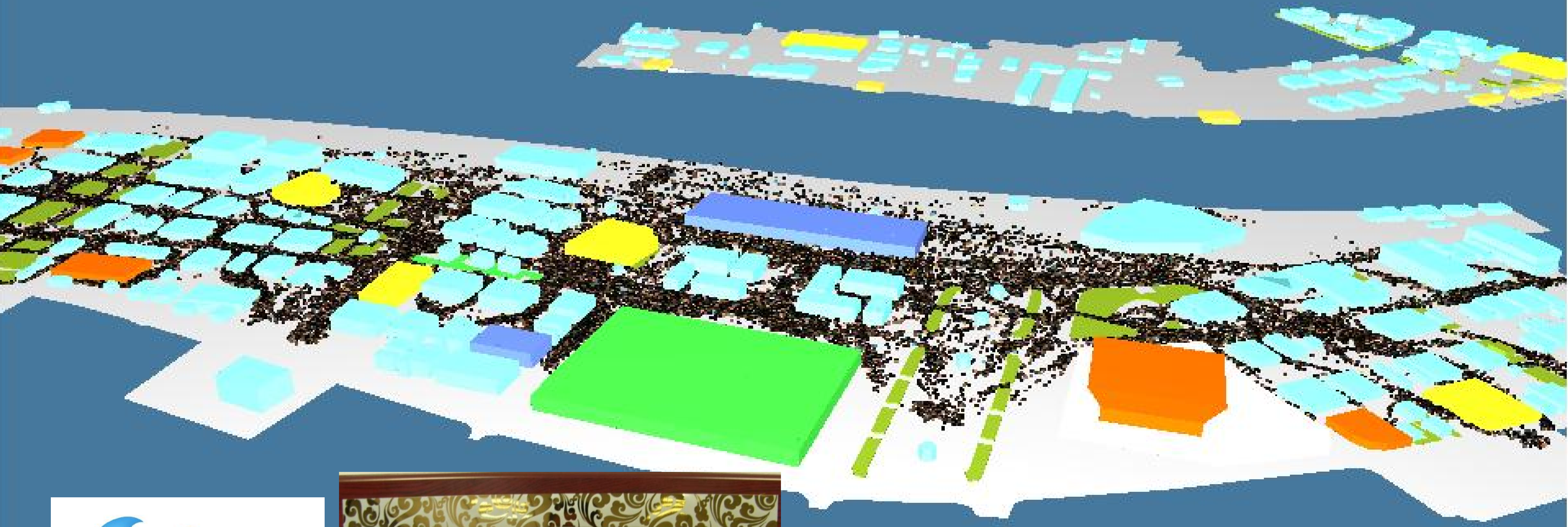

控制台
启动
下载
计算
上传
统计
关闭
退出
动素列表
动作脚本库
动素脚本库
运动库
姿态库
Name
Desc
Path
behavior (6 Items)
bChangeOilFi... behavior\bCh...
bMoveLadder behavior\bMo...
bPullJack behavior\bPu...
bPushJack behavior\bPu...
bUnloadHatch... behavior\bUn...
bUseHammer behavior\bUs...
属性
起始位置 _Param_Pos_StairFront
楼梯顶部 _Param_Pos_StairTopBack
操作位置 _Param_Pos_StairTopForward
引擎开关 _Param_Pos_DoorHandle
油滤安装位置 _Param_Pos_OldFilter
油滤目标位置 _Param_Pos_NewFilter
开始时间 0
持续时间 10
开始时间
务列表


DELMIA V5 - [Process_UpStairs.CATProcess]
开始(S) 文件(F) 编辑(E) 视图(V) 插入(I) 工具(T) 窗口(W) Analyze 帮助(H)
Att
虚拟人维修系统
加载场景
运动合成
运动输出
打开/关闭模拟窗口
DELMIA
Operator21/Assembly_operator2.1预选定



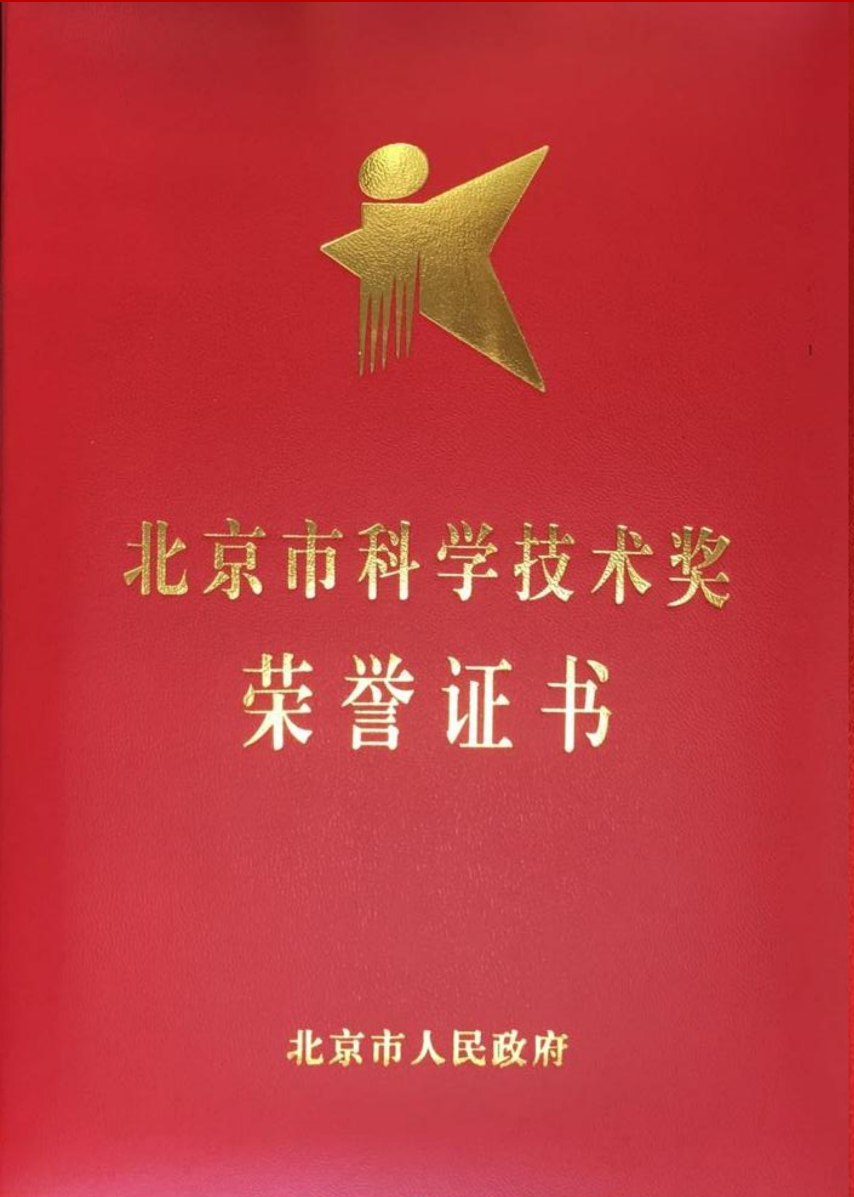
虚拟人维修系统
播放文件
FDA合成
I K反算
路径规划
打开/关闭模拟窗口
运动输出
工具(T)
窗口(W)
Analyze
帮助(H)
P.P.R.
ProcessList
ProductList
ResourcesList
New_Stair_Wheel (New_Stair_Wheel.1)
Stairs (Stairs.1)
Workbench (Workbench.1)
Assembly_operator2 (Assembly_operator2.1)
Manikin (Operator21)
Body
Profiles
Settings
Program
TaskList
Walk close the ladder and push under the aircraft
Climb up the ladder and open the engine cover
task3
Helmet [Electrician_helmet.CATProduct]
Female_worker (Female_worker.1)
Product52 (Movement)
Att
DELMIA
Floor/floor.1/Aircraft.1预选定

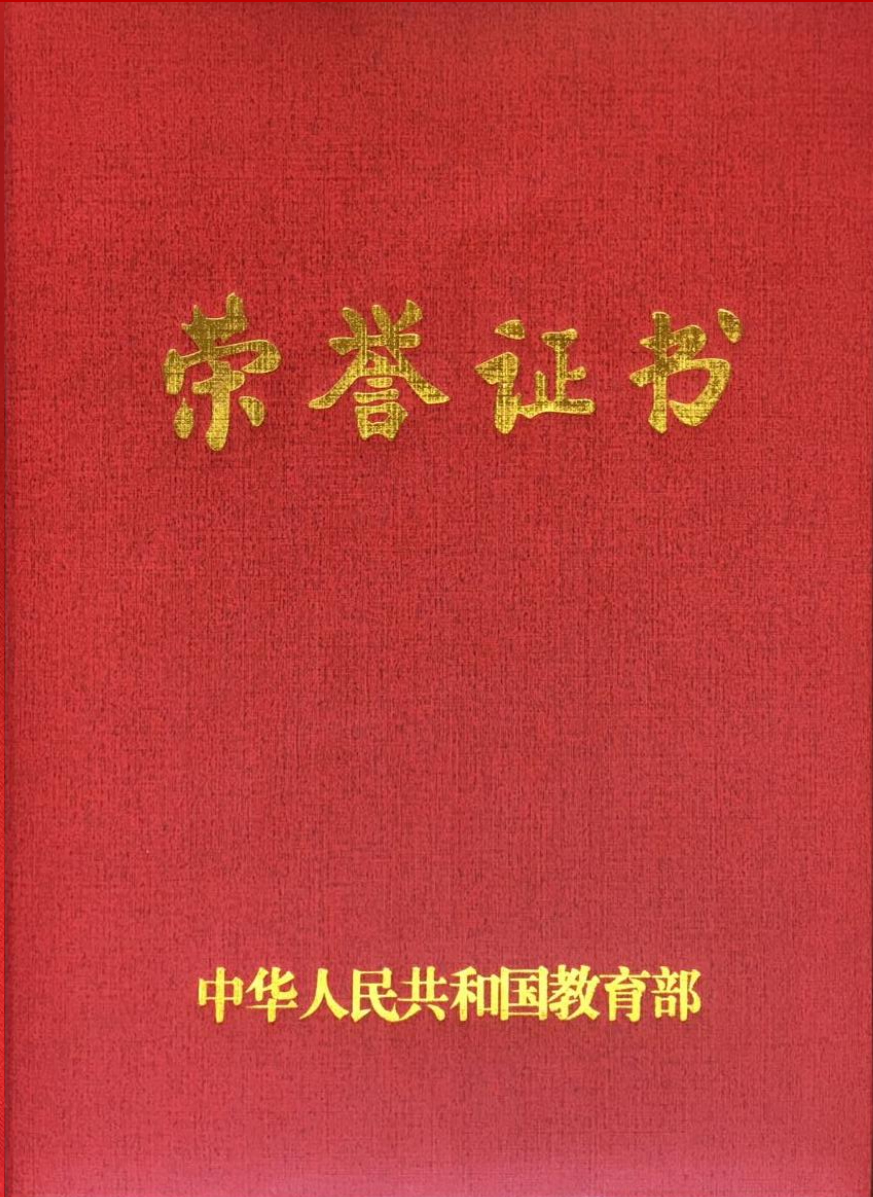





世博科技先进集体
世博科技领导小组
二〇一一年一月

应用评价





北京市科学技术奖

为表彰在推动科学技术进步、对首都经济建设和社会发展作出贡献的集体和个人，特颁此证，以资鼓励。

获奖项目：面向体育训练的三维人体运动模拟与视频分析系统

获奖等级：壹等奖

获奖单位：中国科学院计算技术研究所

北京市人民政府

二〇〇八年十二月

证书

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奖励日期：2012 年 11 月

证 书 号：2012921227

总 装 备 部

为表彰在促进科学技术进步工作中做出重大贡献，特颁发此证书。

获奖项目：人体运动捕获数据处理及重用技术

获 奖 者：夏时洪(第2完成人)

奖励等级：技术发明奖二等奖

奖励日期：2019年1月

证 书 号：2018-176

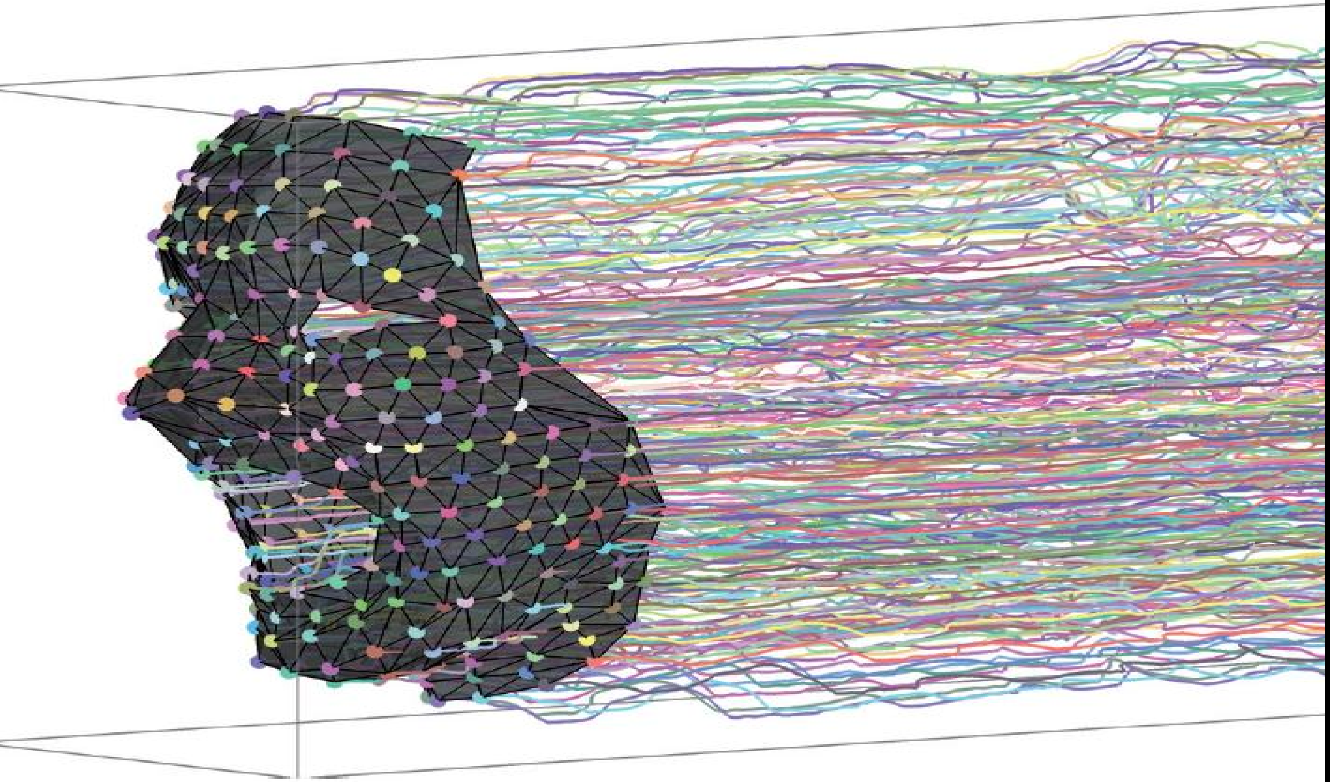
中华人民共和国教育部

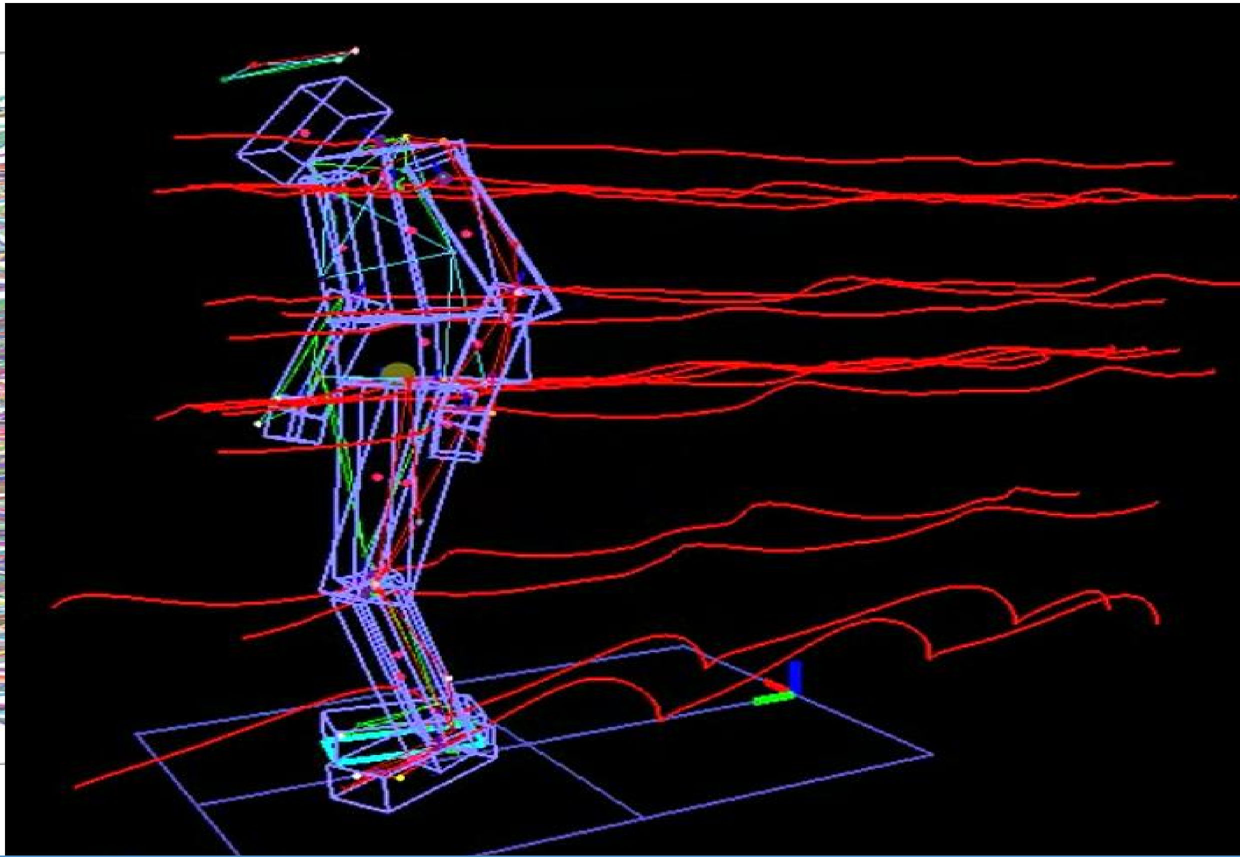
教育部

二〇一九年一月十五日

Part 2: 代表性基础研究

基础问题





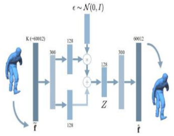
如何用尽可能少的（控制）参数建模人体运动系统的高维、动态时空信息？

研究挑战

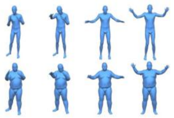
1. 人体运动建模仿真结果符合人的视觉认知吗？　*（“诡异谷”问题）*
2. 人体运动建模仿真结果符合牛顿力学/动力学原理吗？符合物理现实吗？　*（“物理/现实真实感”问题）*
3. …………

基础交叉前沿学术研究(2015至今)

geometry learning

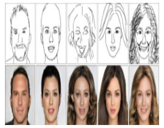


deep deformation analysis
AAAI/CVPR 2018



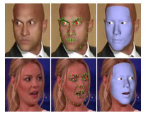
deep deformation transfer
SIGGRAPH Asia 2018

deep deformable mesh
SIGGRAPH Asia 2019



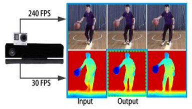
DeepFaceDrawing
SIGGRAPH 2020

motion+geometry



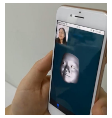
face/eyegaze mocap
SIGGRAPH 2016

shape deformation
ACM TOG 2016



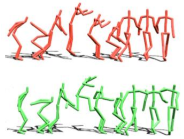
fast motion capture
IEEE TVCG 2019

sparse mesh deformation
IEEE TVCG 2020

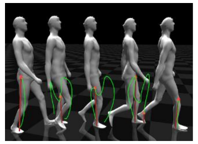


deep eyegaze capture
IEEE TVCG 2021

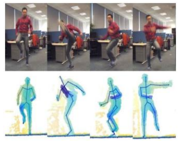
motion



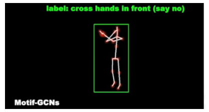
motion style
SIGGRAPH 2015



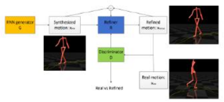
data-driven inverse dynamics
SIGGRAPH Asia 2016



3D pose estimation
IEEE VR 2018



deep motion recognition
AAAI 2019



deep motion synthesis
IEEE TVCG 2021

Neural3Points
SCA 2022

SIGGRAPH论文0→1

ACM TOG/IEEE TVCG论文1→10

成果一：高精度人体运动风格建模

为了准确复现多种多样的人体运动，需要合成风格可调的人体运动。

关键科学问题：如何构建人体运动风格特征，建模外部约束同运动风格的最优映射关系？

项目资助：

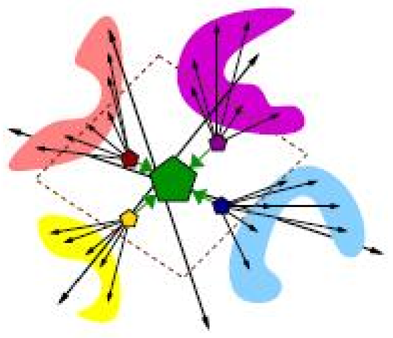
国家自然科学基金，

风格化人体运动合成新方法研究，61173055；

国家科技支撑计划，

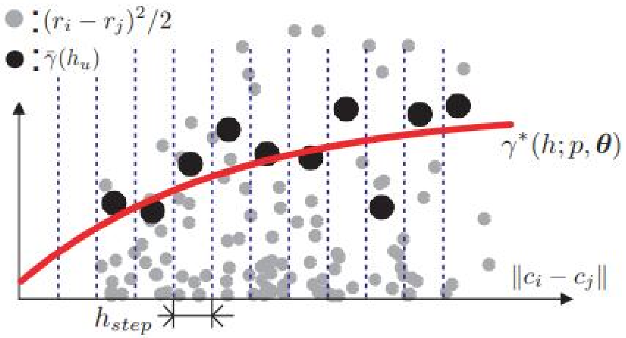
国家应急演练仿真服务平台及应用示范，2013BAK03B07。

现状与挑战



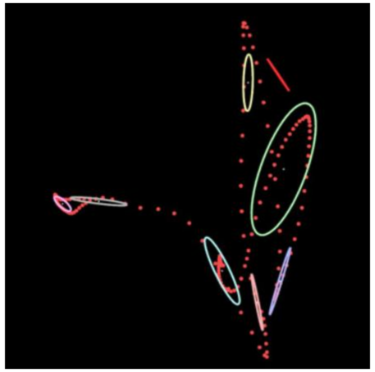
stylistic hidden Markov model

SIGGRAPH 2000



universal kriging model

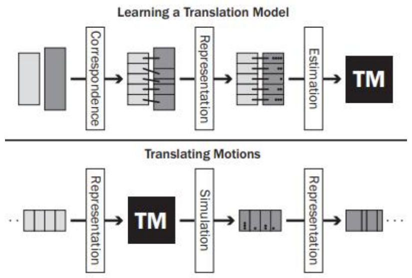
SIGGRAPH 2005



Gaussian process model

SIGGRAPH 2004/ICML 2007/ACM TOG 2009

$$\mathbf{x}_{t+1} = \mathbf{A}\mathbf{x}_t + \mathbf{B}\mathbf{u}_t + \mathbf{e}_t$$
$$\mathbf{y}_t = \mathbf{C}\mathbf{x}_t + \mathbf{D}\mathbf{u}_t + \mathbf{f}_t.$$



linear time-invariant (LTI) model

SIGGRAPH 2005

实时合成风格可调的人体运动仍然存在以下挑战：

1）不同运动之间的时空映射关系异常复杂，难以准确建模运动风格；

2）人体运动中动作与运动风格的识别、理解是公开难题；

3）运动风格可调性不高，难实时交互。

关键思路

在线构建局部**自回归模型**,准确建模运动风格

$$y_t = u_t^T \beta_t + \varepsilon_t$$

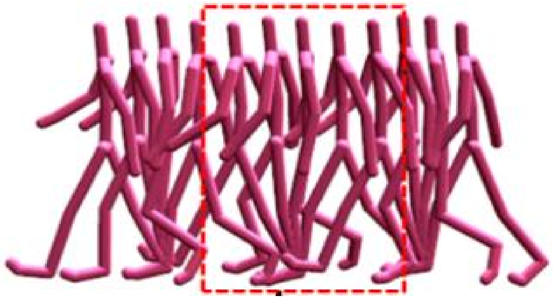
t时刻输出

回归变量

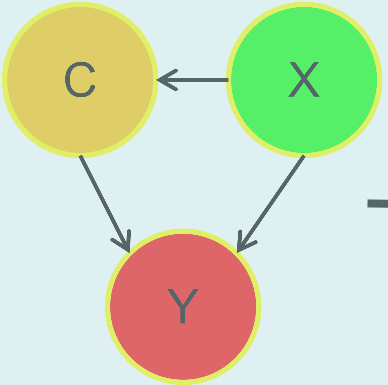
$(x_{t-1}, y_{t-1}, x_t, 1)^T$

回归模型参数

$(a_t, b_t, c_t, d_t\ )^T$



构建**混合自回归模型**，处理未标注运动



混合模型的概率图表示

$$\begin{cases} f(y_t|u_t) = \sum_{j=1}^{c} \lambda_j \phi(y_t|u_t^T \beta_t^j, \sigma_j^2) \\ \lambda_j \propto \exp(-\frac{D_j}{\delta \min_j D_j}), j = 1, \dots, c \end{cases}$$

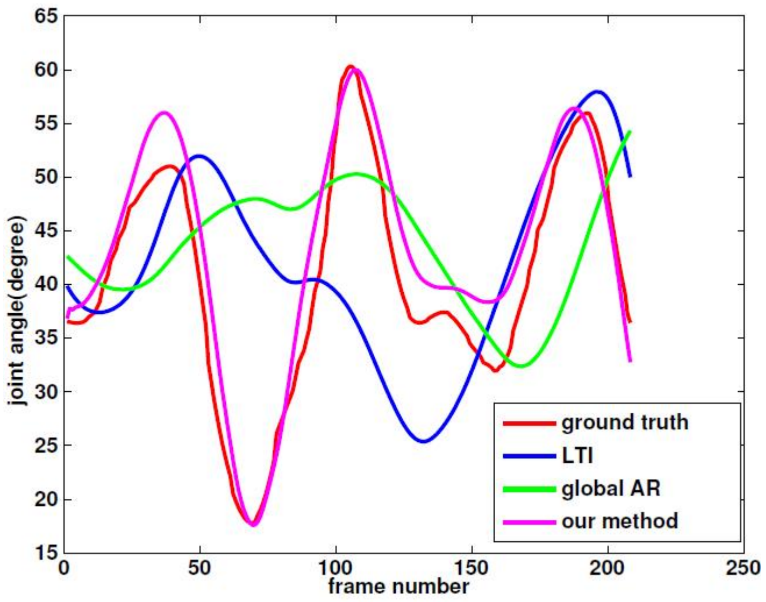
高斯分布

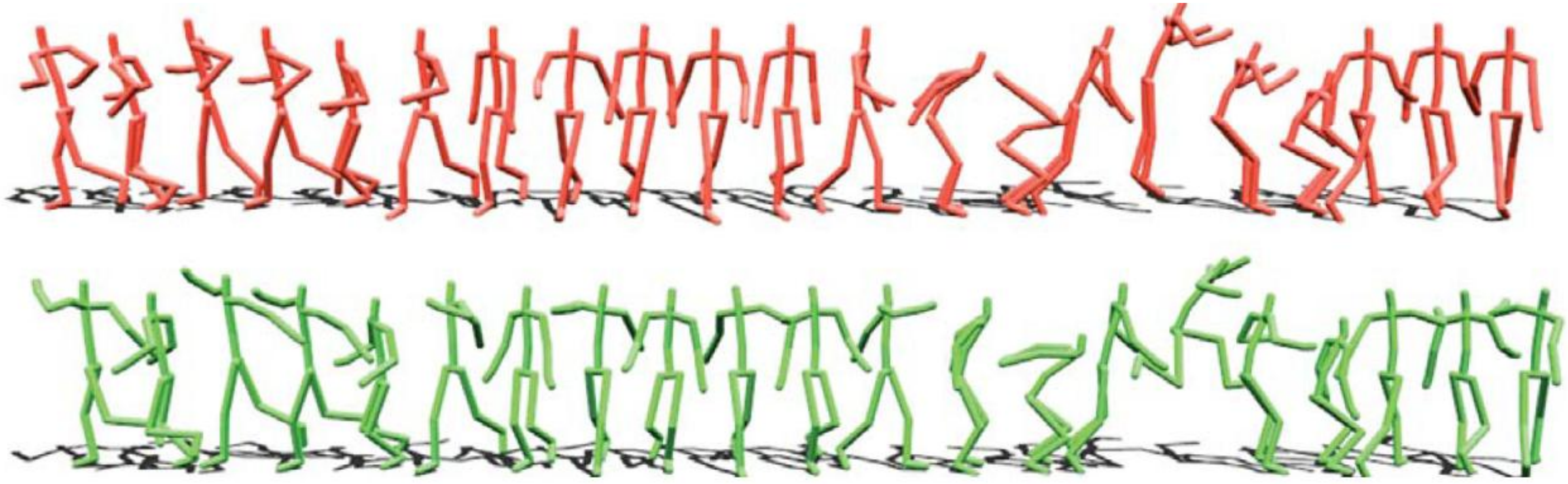
KNN平均距离

$$E(y_t) = \sum_{j=1}^{c} \lambda_j u_t^T \beta_t^j$$

创新点一：提出了运动风格的局部混合自回归模型，有效解决了高精度运动风格建模问题。

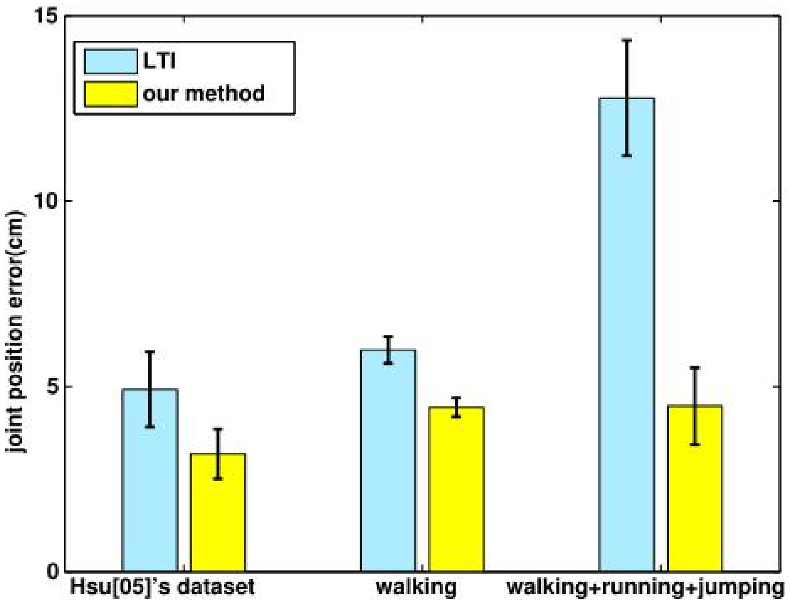
1. 揭示了运动风格全局模型的缺陷，国际首次实现了异质人体运动风格迁移、能够实时在线地合成风格可调的自然人体运动

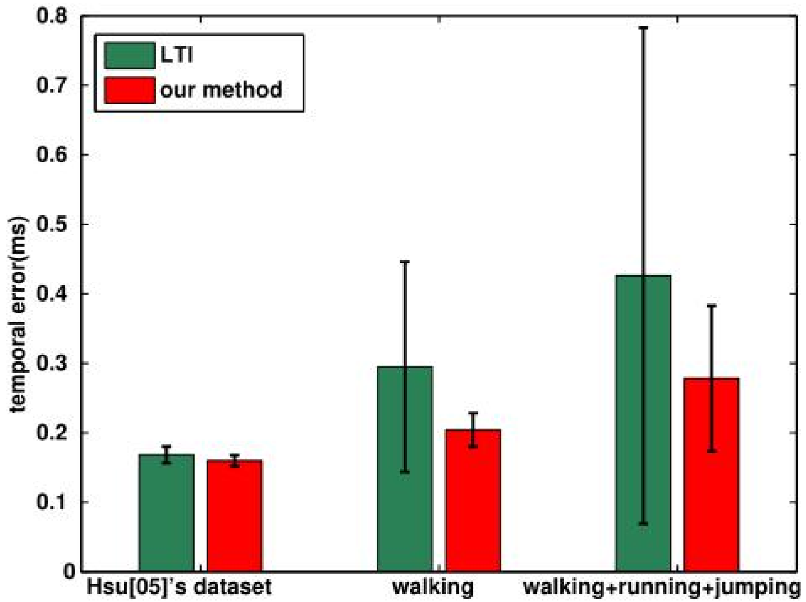




创新点一：提出了运动风格的局部混合自回归模型，有效解决了高精度运动风格建模问题。

1. 揭示了运动风格全局模型的缺陷，国际首次实现了异质人体运动风格迁移、能够实时在线地合成风格可调的自然人体运动
2. 同质运动风格合成误差相比MIT提出的LTI模型降低了35.45%





左：关节角重建误差；右：时间重建误差

同质运动风格合成误差相比MIT提出的LTI模型降低了35.45%

Paper ID: 0565

成果二：高精度人体运动动力学建模

为了准确复现人体运动的动态调控过程，需要建立高精度的人体运动动力学模型。

关键科学问题：如何校准模型参数，构建动力学先验模型求解生物力学准确的接触力？

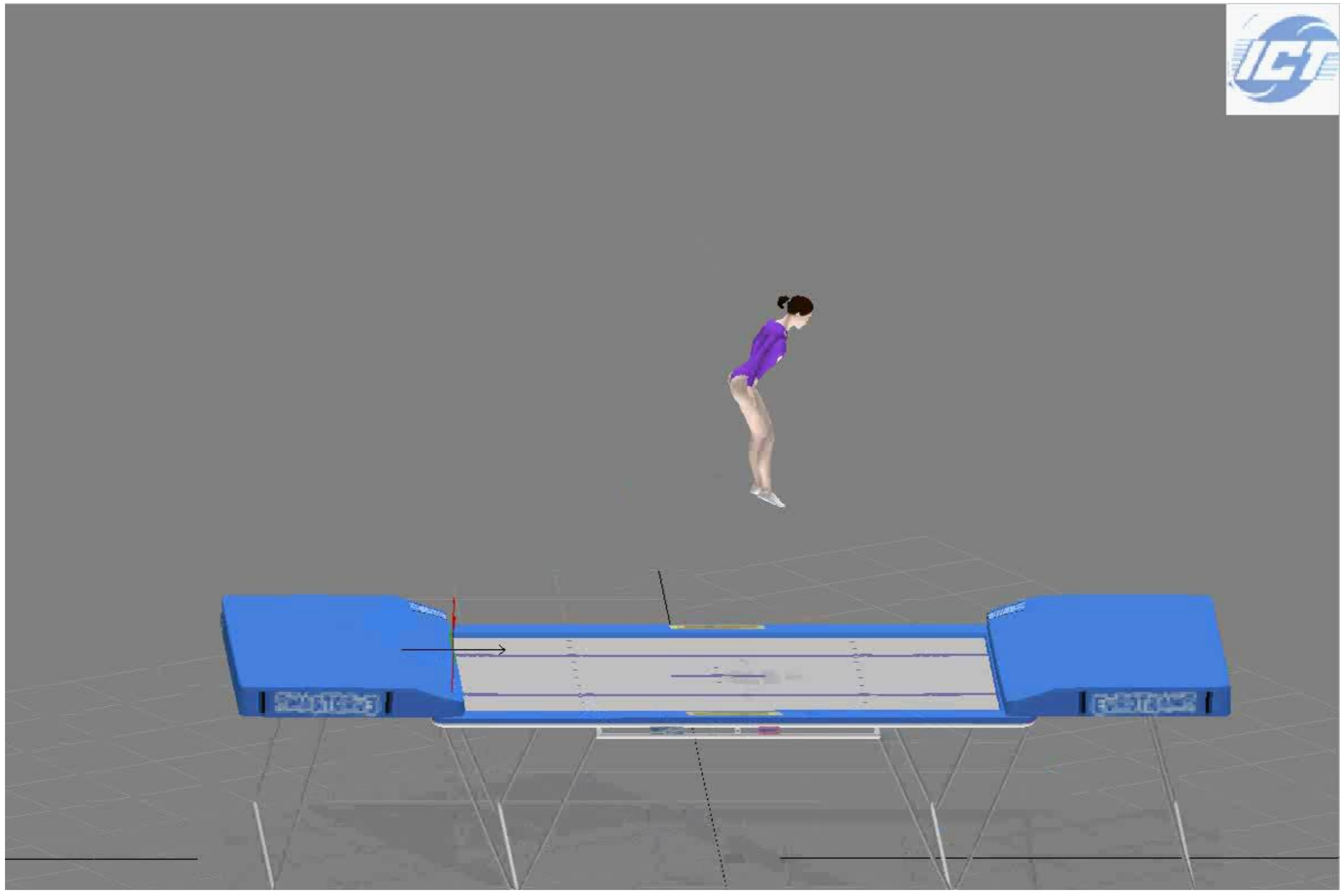
项目资助：

国家863课题，

虚拟环境中基于物理特性的虚拟角色运动行为建模，2007AA01Z320；

国家科技支撑计划，

面向医疗康复的三维肢体运动功能分析系统研制，2008BAI50B07。



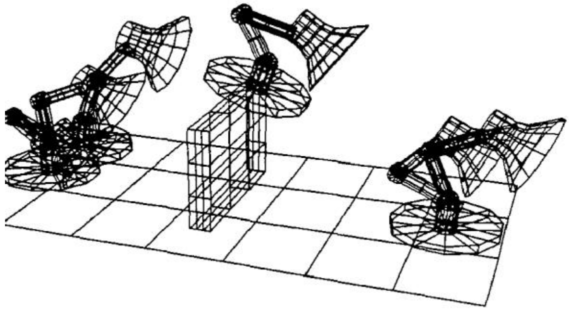
现状与挑战

$$\min_{\mathbf{X}} G(\mathbf{X}) = \sum_{t=0}^{n-1} ||\mathbf{f}(t)||^2$$

$$\text{subject to} \quad \mathbf{C}_i(\mathbf{X}), i = 0 \ldots n-1$$

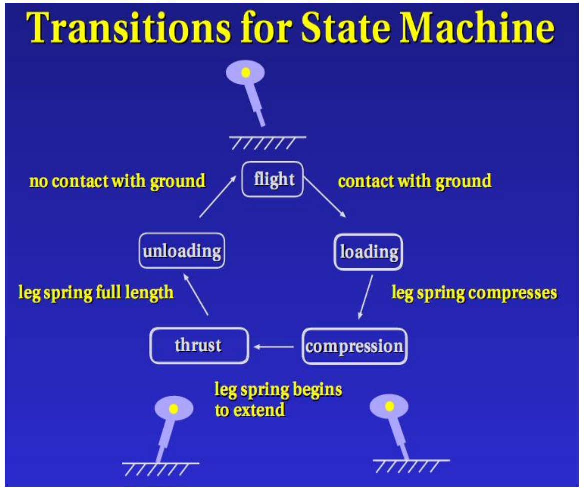
$$\mathbf{q}_0 = \mathbf{q}_a$$

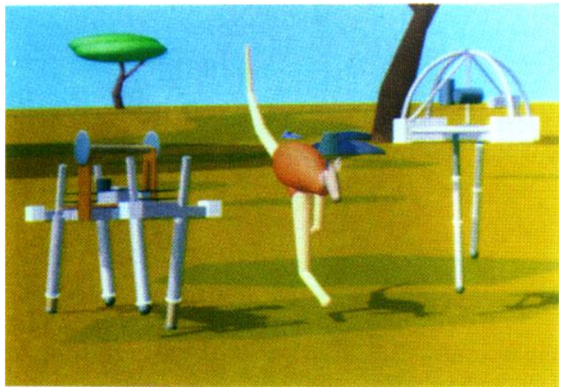
$$\mathbf{q}_{n-1} = \mathbf{q}_b$$



Spacetime constraints / trajectory optimization

SIGGRAPH 1988， 1992， 1999， 2009





$$F = k_d{}^*(\theta_d - \theta) + k_v{}^*(\dot{\theta}_d - \dot{\theta})$$

finite state machine / Proportional-Derivative controller

SIGGRAPH 1991， 1995， 2007， 2008， 2012； SIGGRAPH Asia 2009

动力学模型参数以及逆向动力学求解的准确性问题还存在诸多技术难题：

1）与动力学方程相关的动态人体肢体质量和质心等人体惯性参数是个体相关的，很难测量。

2）人体与地面接触的双支撑阶段，逆向动力学问题是歧义的。

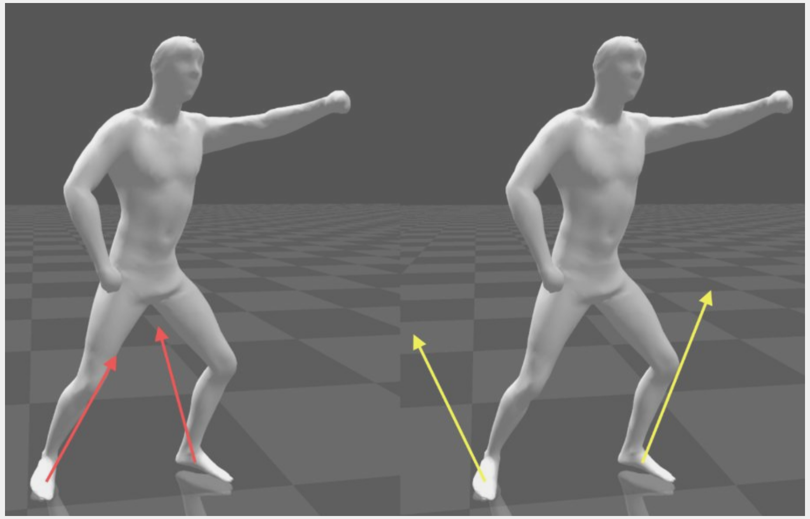
3）人体运动逆向动力学求解过程是与人体惯性参数耦合在一起的。

关键思路

建立人体姿态、人体与地面接触力/压力及接触约束、人体运动动力学约束一体化的动力学模型

采用变量分步迭代优化技术实现动力学模型参数的校准，利用数据驱动的思想求解逆向动力学问题

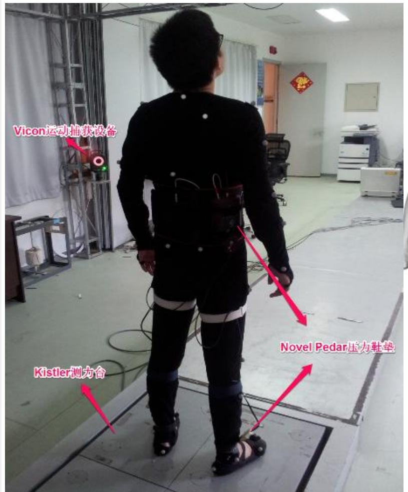
逆向动力学问题是一个欠定的问题

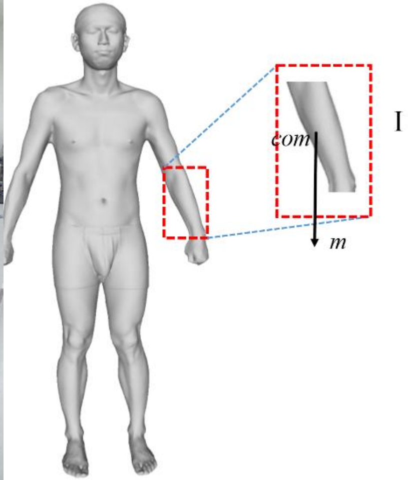


人体根关节受力和力矩为0，约束为6维；双支撑时期未知变量为左右脚接触信息，共计12维。欠定。

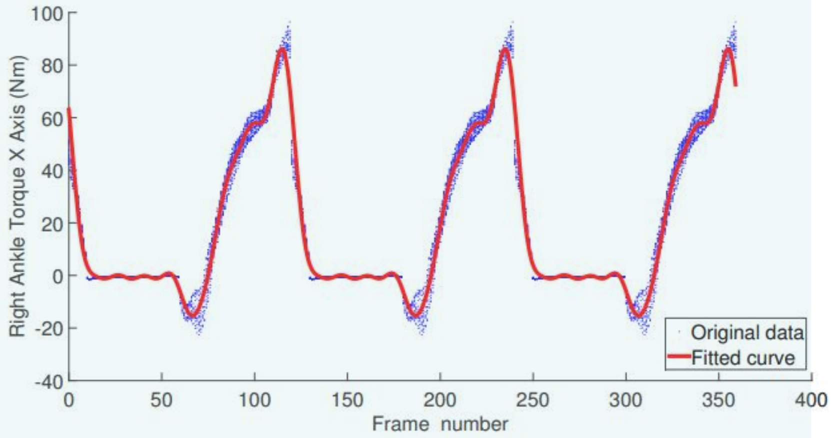
$$\min_{m,com,I} w_1 \sum_{i=1}^{K} \|(M(q)\ddot{q}+C(q,\dot{q})+G(q)-J_F^T F-J_{\tau_z}^T \tau_z)_{1:6}\|^2$$
$$+w_2 \sum_{j=1}^{n} \|I_j - I_{j\text{init}}\|^2 + w_3 \sum_{j=1}^{n} \|m_j - m_{j\text{init}}\|^2$$
$$+w_4 \sum_{j=1}^{n} \|com_j - com_{j\text{init}}\|^2$$

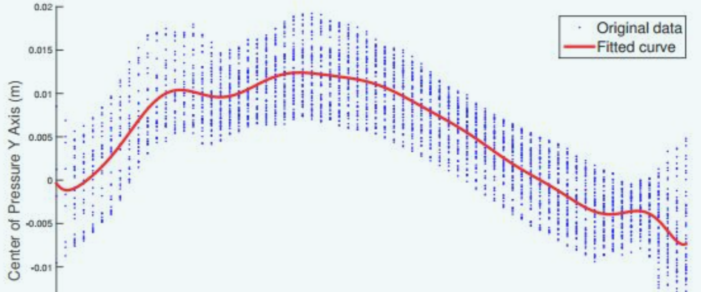
$$\text{subject to } m_{\text{total}} = \sum_{j=1}^{n} m_j.$$





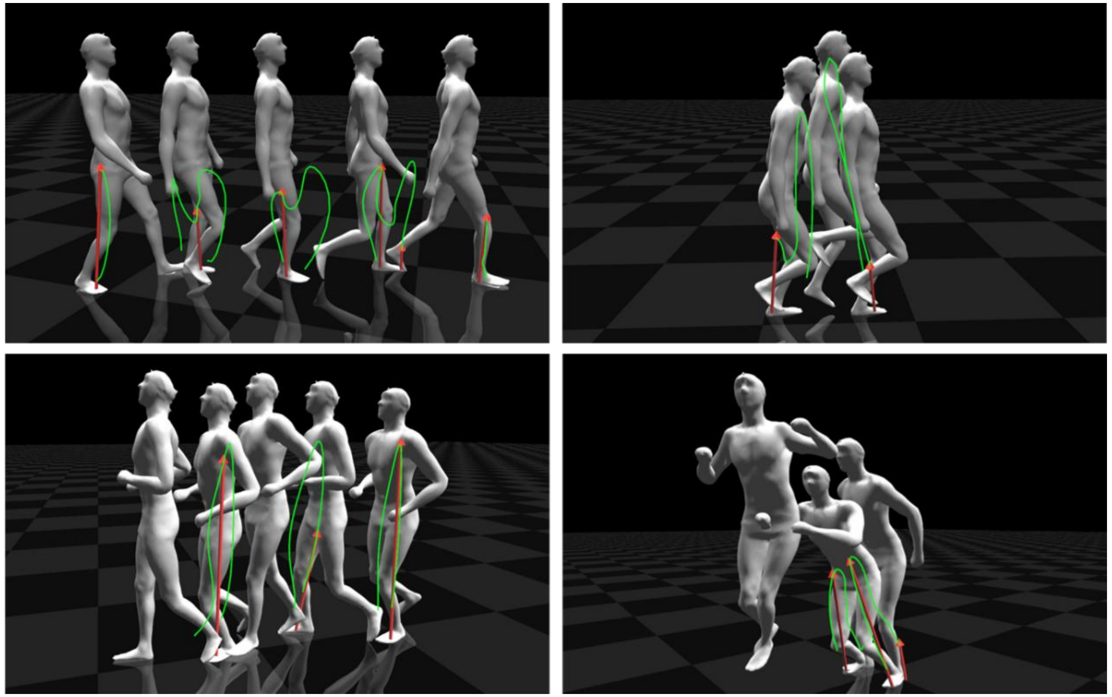
$$\min_{F,cop,\tau_z} w_1 \|(M(q)\ddot{q}+C(q,\dot{q})+G(q)-J_F^T F-J_{\tau_z}^T \tau_z)_{1:6}\|^2$$
$$+w_2\|F_z - F_{z\text{init}}\|^2 + w_3\|cop - cop_{\text{init}}\|^2$$



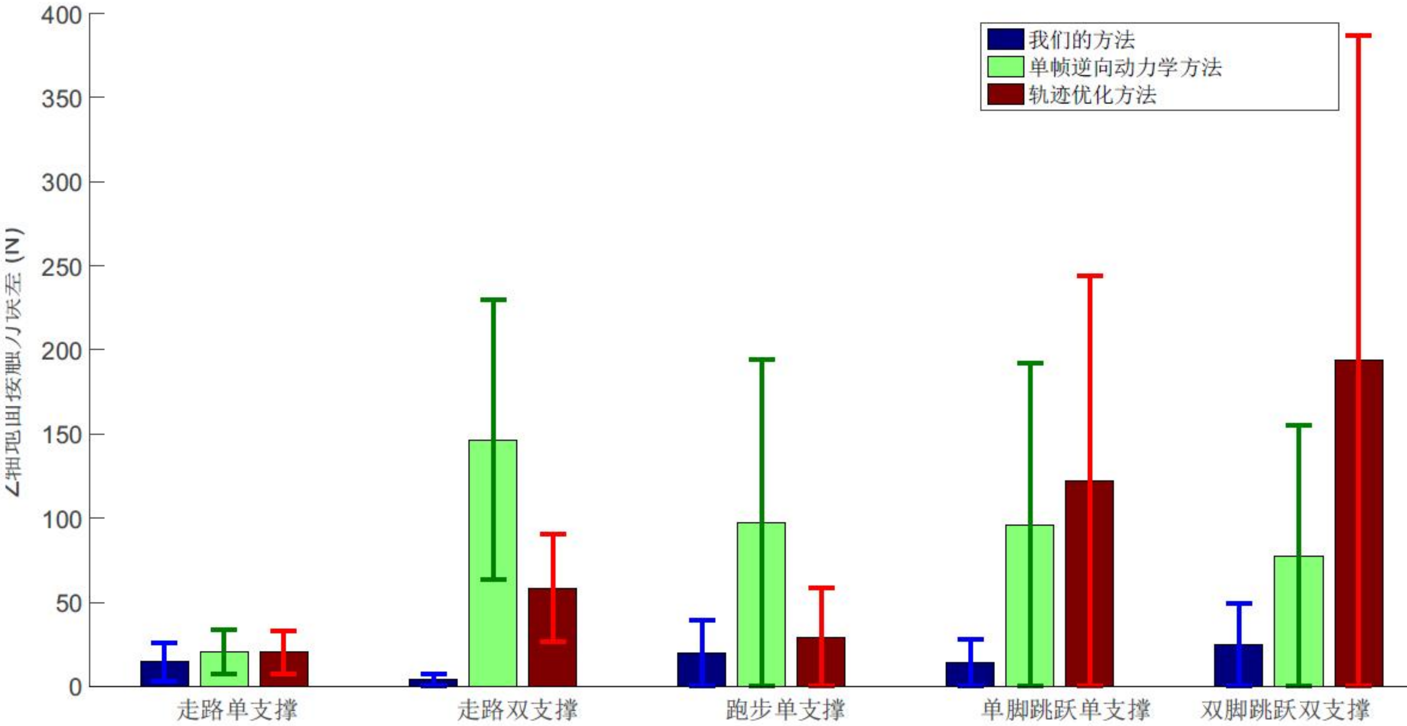


创新点二：提出了个体化人体运动动力学模型，有效解决了高精度动力学建模问题。

1. 揭示了人体运动逆向动力学问题的歧义现象，国际首次实现了生物力学准确的逆向动力学求解、高精度地复现自然人体运动的动态调控过程
2. 竖直方向地面接触力重建误差相比经典的轨迹优化方法降低了92.63%



多种运动类型的逆向动力学求解

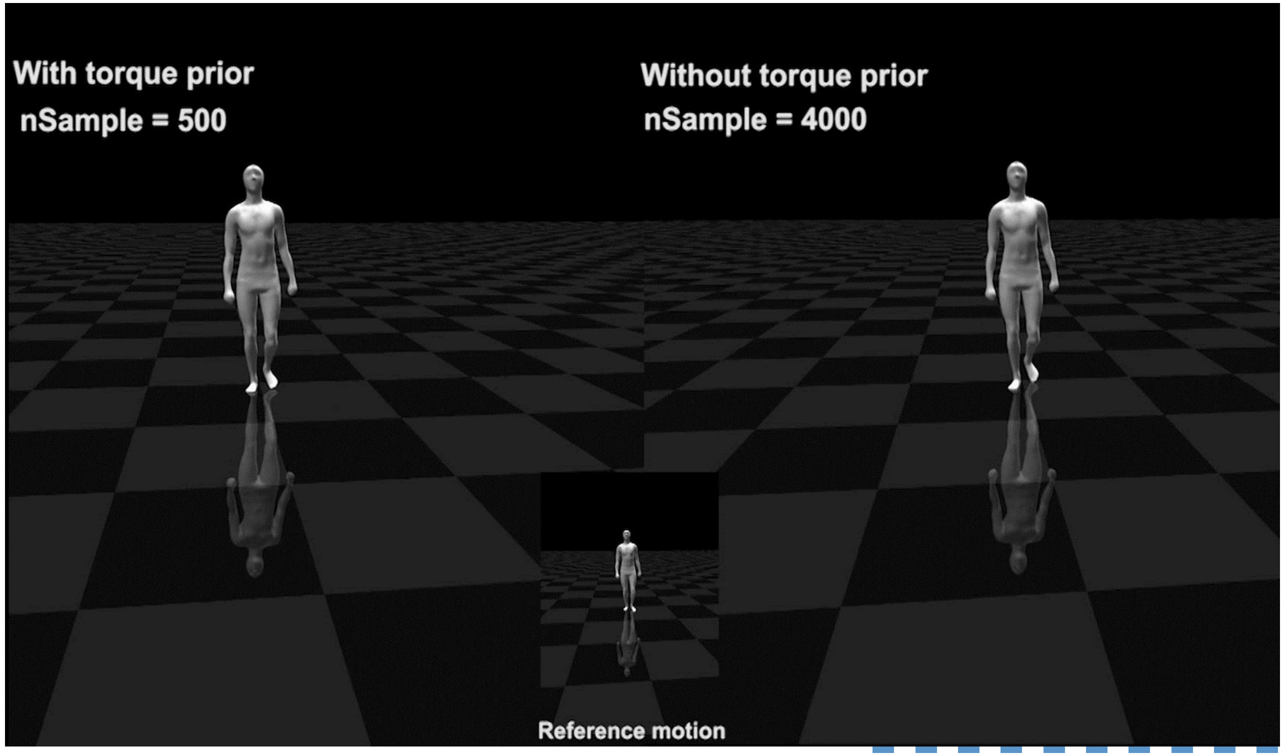


竖直方向地面接触力重建误差

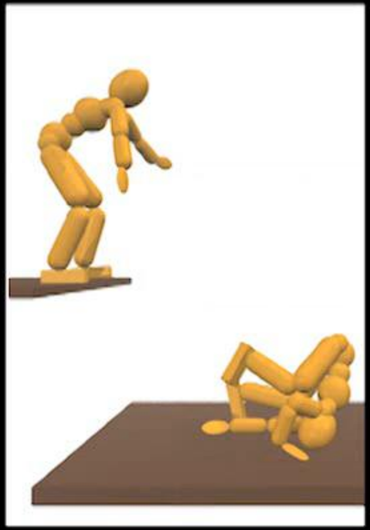
创新点二：提出了个体化人体运动动力学模型，有效解决了高精度动力学建模问题。

1. 揭示了人体运动逆向动力学问题的歧义现象，国际首次实现了生物力学准确的逆向动力学求解、高精度地复现自然人体运动的动态调控过程
2. 竖直方向地面接触力重建误差相比经典的轨迹优化方法降低了92.63%
3. 将基于随机采样的多接触点运动控制方法的随机采样点数减少了87.5%

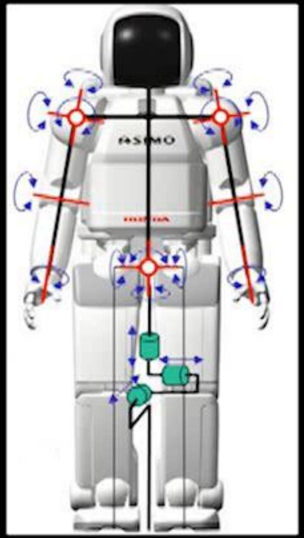
用于一种有限状态机控制模型（Sampling-based contact-rich motion control，SIGGRAPH 2010），500个随机采样点即可达到原工作需要4000个随机采样点的运动合成效果，即，随机采样点数降低为原来的1/8，减少了87.5%。



Inverse dynamics is a core problem in



Character Animation



Robotics

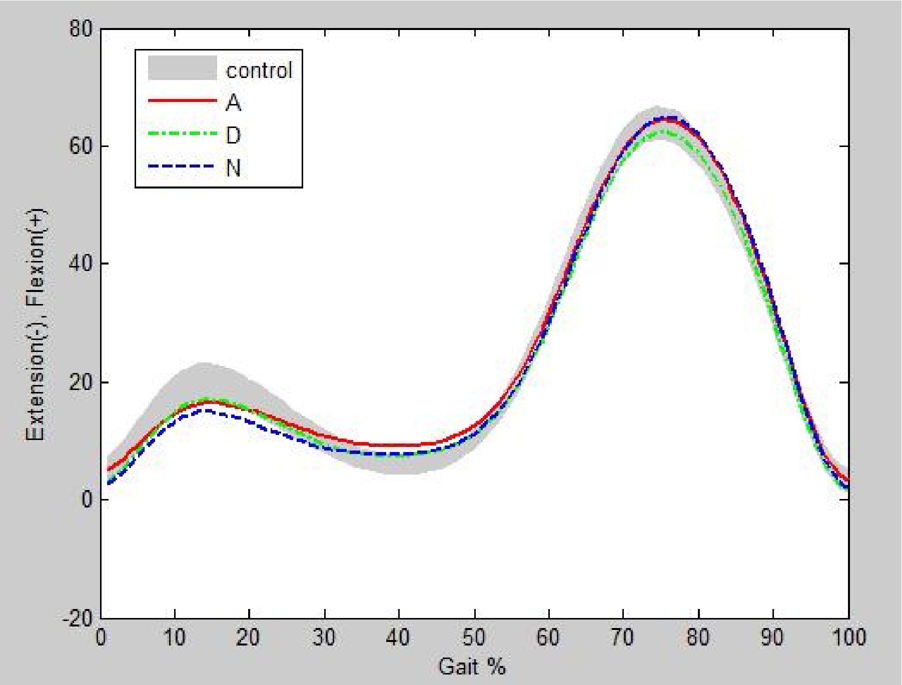
Biomechanics

运动医学中的术式评估

如何评估ACL(前交叉韧带断裂)不同术式的效果?

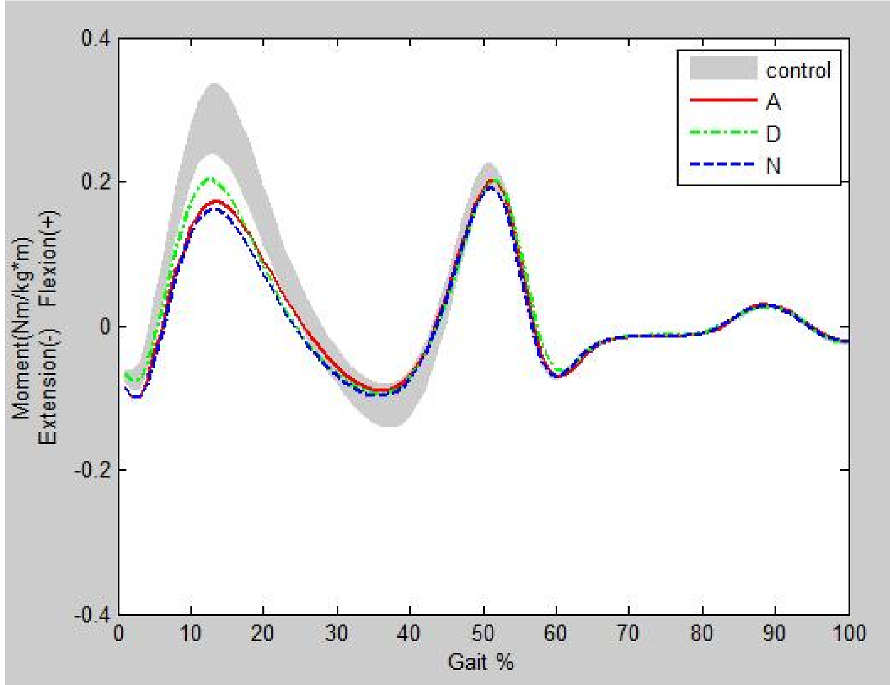
为什么ACL(前交叉韧带断裂)术后会二次受伤/病变（膝骨关节炎）？

北医三院运动医学研究所提供15个ACL重建病人（术后两年）的正常步行运动捕获数据



左图：运动学分析

右图：动力学分析



成果三：高精度人体运动几何形状建模

为了感知用户的运动行为，需要从图像视频中建立高精度的运动人体几何模型。

关键科学问题：如何构建先验知识，用人体运动计算模型精准重建运动人体几何形状？

项目资助：

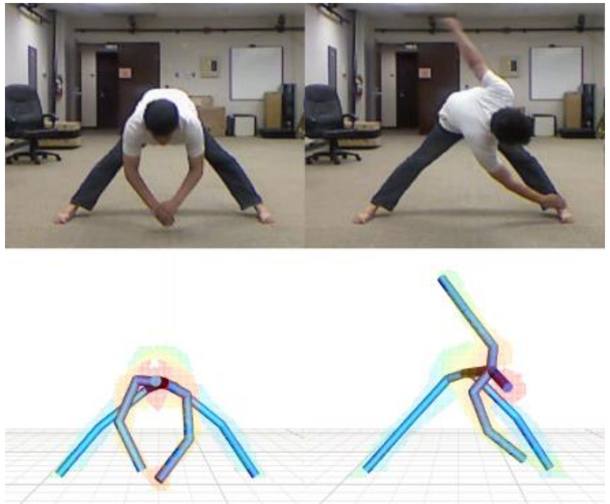
国家自然科学基金，

基于单目RGB/RGBD相机的身体运动和面部运动同步捕获方法研究，61772499；

国家863课题，

3D内容视觉获取技术及设备，2012AA011501。

现状与挑战



3D pose estimation / tracking

SIGGRAPH 2005，SIGGRAPH Asia 2012

3D pose + shape

SIGGRAPH 2005, 2016

face expression + shape

SIGGRAPH 2007, 2013

基于单目图像视频的运动人体几何建模研究，仍然存在许多技术挑战：

1）从单目图像视频恢复三维结构信息本质是一个病态问题。

2）图像视频存在光照变化，并且其中的运动人体也存在遮挡和自遮挡等因素。

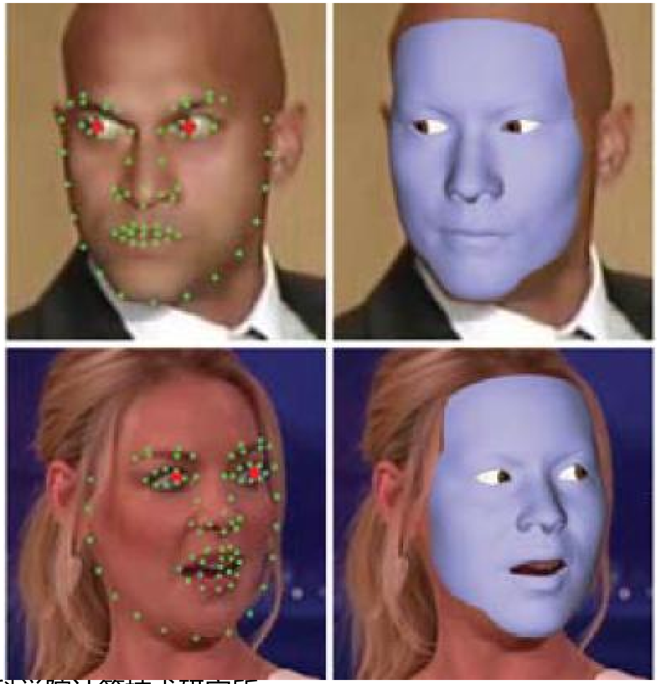
3）运动人体的姿态和几何形状是耦合在一起的。

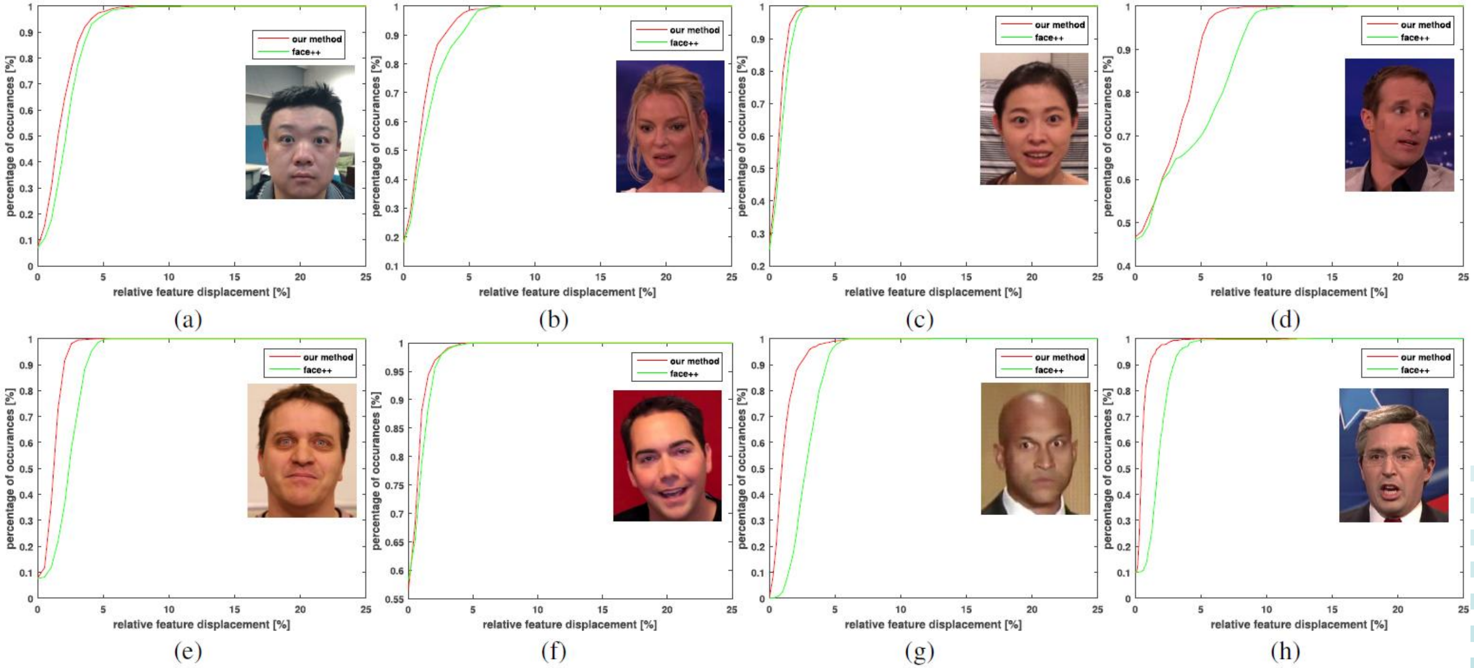
关键思路

充分利用身体姿态与运动、面部表情与形状的先验知识恢复重建运动人体的几何形状和运动信息

创新点三：提出了基于运动模型的运动人体姿态、表情和几何模型重建框架，较好解决了高精度运动人体几何建模问题。

1. 能够从单目RGB-D图像视频一体化重建运动人体三维姿态/几何形状、表情和眼球运动
2. 从单目深度视频实时在线重建出合理、准确的3D人体运动姿态序列，平均帧率约为100fps
3. 揭示了运动人体大尺度几何形变现象，提出的数据驱动的非刚体模型三维重建方法优于经典的SCAPE方法
4. 国际首次从单目彩色视频实时重建3D人脸表情、头部姿势以及眼球运动

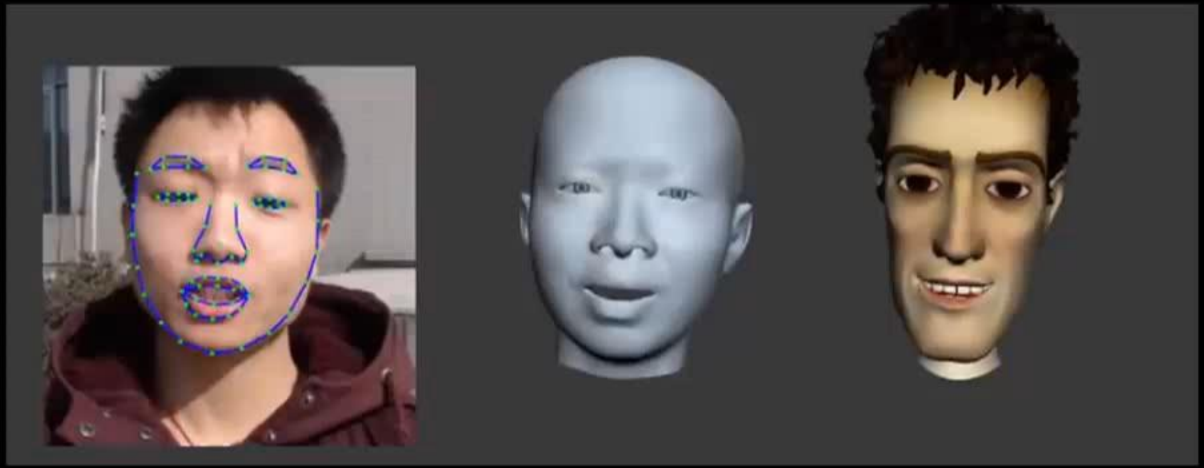




Facial performance capture is a hot topic

[Garrido et al. 2014]

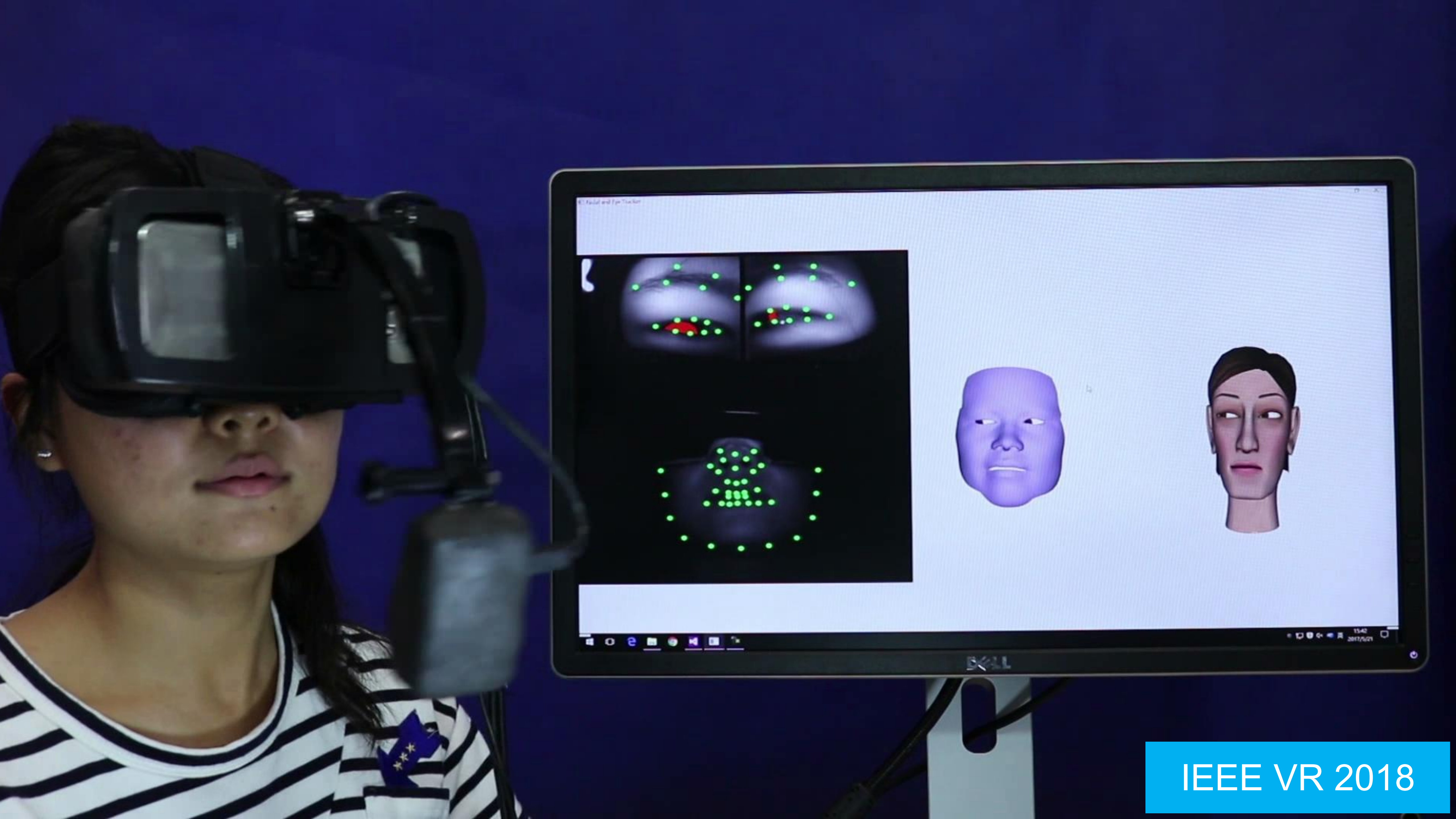
[Huang et al. 2014]



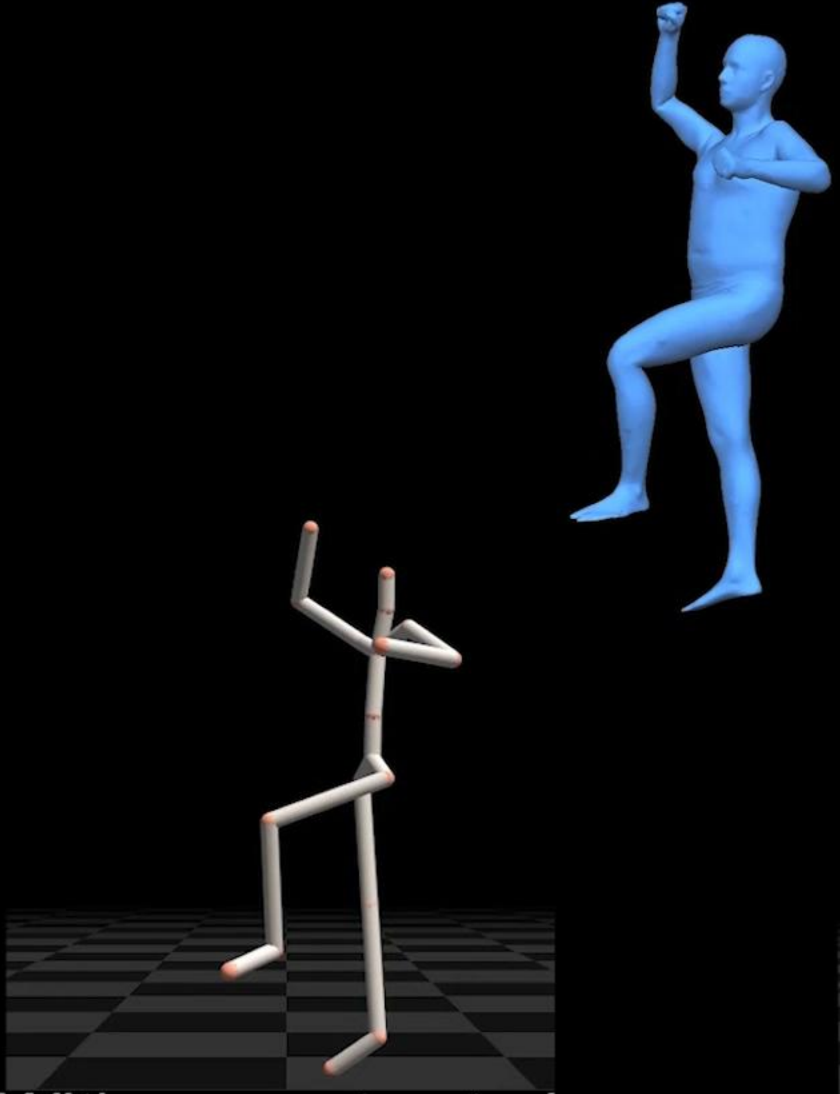
[Cao et al. 2014]

[Shi et al. 2014]

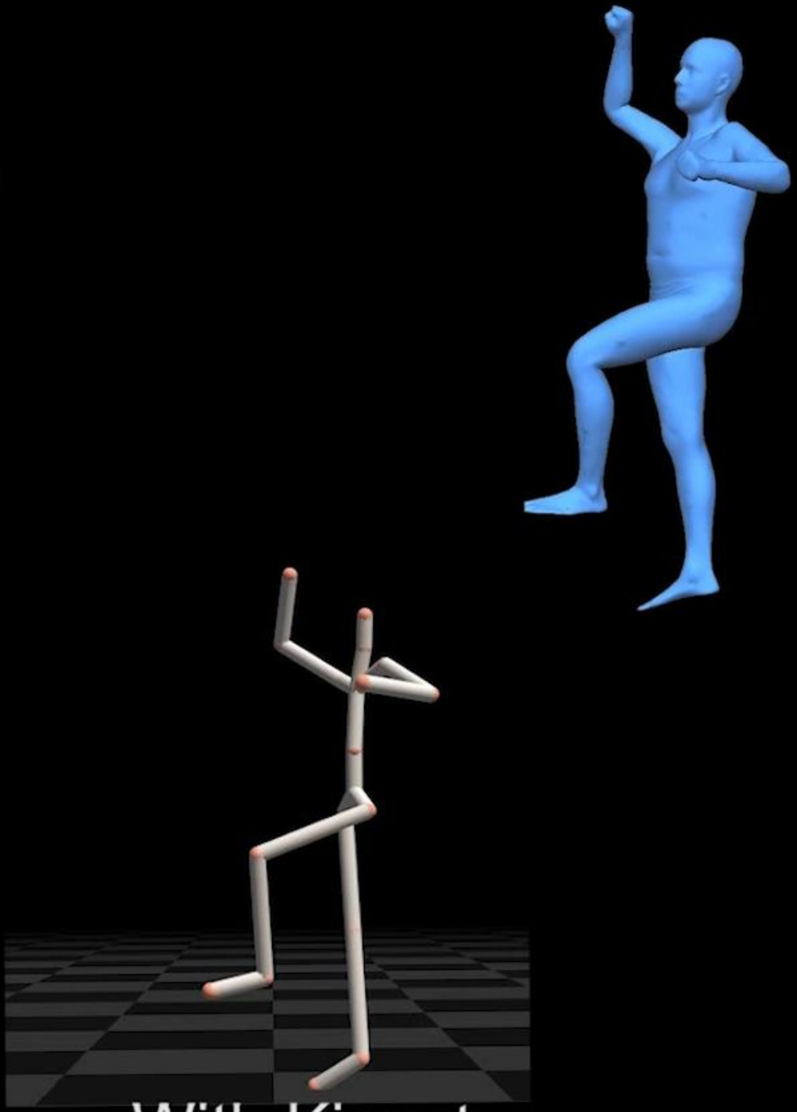

Eye gaze heat map


IEEE VR 2018

RGB



With reconstructed
high-frame-rate (240fps)
depth maps



With Kinect-
captured (30fps)
depth maps

Playback Speed: 0.1x (24fps)

研究贡献

施引文献的研究领域分布广泛：计算机图形学与交互式技术，计算机视觉与图像处理，计算机动画与虚拟世界，虚拟现实，机器人，交互式智能系统，信号处理，应用感知，计算机辅助几何设计，虚拟康复治疗，情感陪护机器人，情感合成与识别等。

学会办公室联系电话：010-82317098　82310612。

特此公示

中国仿真学会

2017 年 7 月 21 日

2017 年度中国仿真学会科学技术奖拟授奖项目公示名单

（按得票数排序，同票数按答辩顺序排序）

获奖等级	奖励类别	项目名称	推荐单位	主要完成人
一等奖	科技进步类	数据驱动的人体运动建模仿真技术	中国科学院计算技术研究所	夏时洪　高　林　王　涵
	自然科学类	大型复杂装备协同式虚拟维修训练系统基础技术研究	中国人民解放军火箭军工程大学	张志利　李向阳　高钦和 梁　丰　王　锐　曹继平 侯宽耀　谭立龙　甄占昌 龙　勇　汤志波　管文良 佟　昭　魏玉淼　邓刚锋
	自然科学类	基于行为的交通仿真模型	清华大学	吴建平　杜怡曼　黄　玲 贾宇涵　朱春丽　郑少将 李婷婷　许　明　孔　源 胡冬梅　胡可臻　杨森焱 刘明宇
二等奖	科技进步类	可扩展仿真平台	北京华如科技股份有限公司	张　柯　闫　飞　王山平 陈敏杰　谭　雄　王　玮 王　军　涂　智　孙　昊
	科技进步类	半实物仿真实时光纤通讯网络	中国航天科工集团第三总体设计部	王宪宗　姚　昱　郭卓锋 李艳雷　朱震宇　刘　鑫 孔文华　史　航　周莉莉


中国计算机学会
CCF

中国计算机学会
2018年科学技术奖
技术发明一等奖

授予
中国科学院计算技术研究所
“三维人体运动建模关键技术及应用”项目

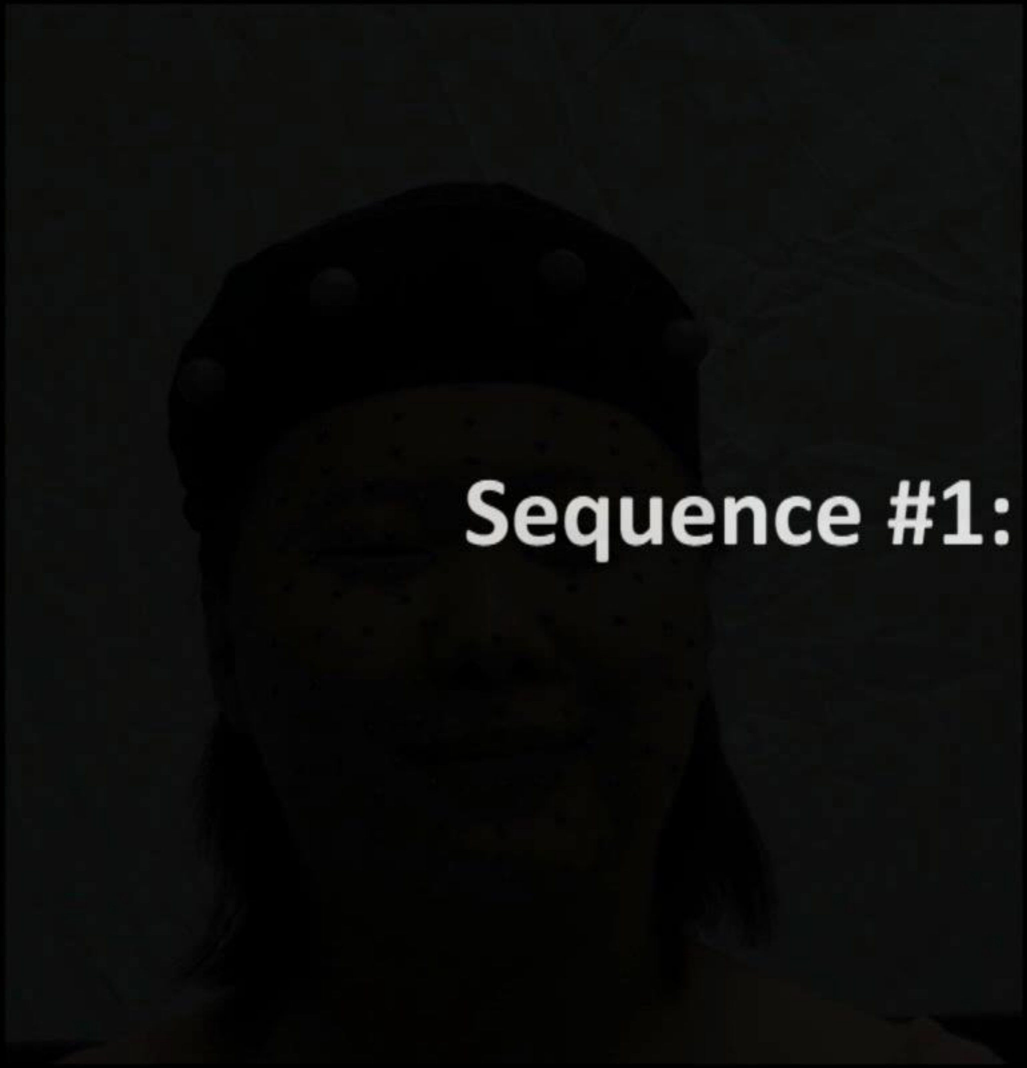
主要完成人 夏时洪、高林、魏毅、苏乐、
邓小明、袁铭择

理事长　　　　　　　　奖励委员会主席

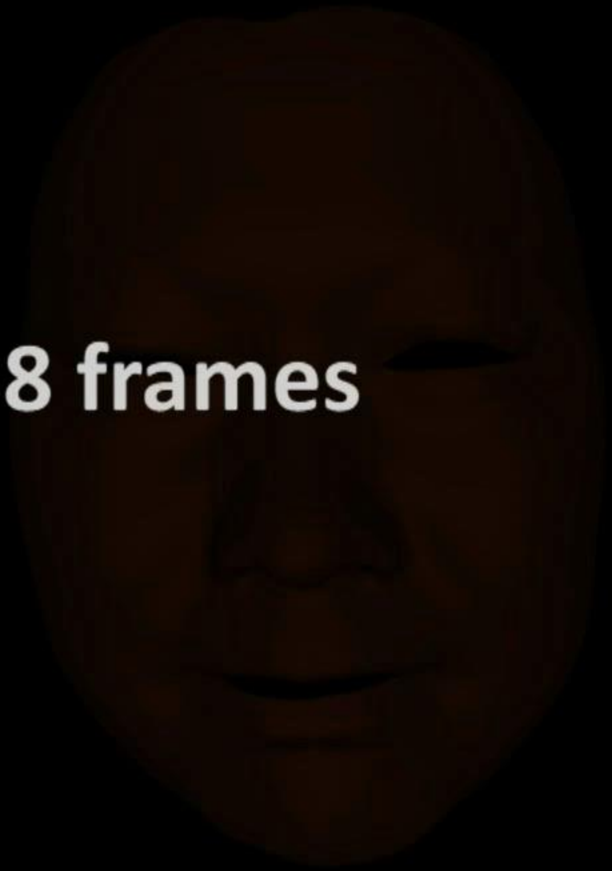
中国计算机学会
2018年10月26日

编号：CCF-AWD-18-013

代表性引用与应用



Sequence #1: 8758 frames



代表性引用与评价

Hao Li, Thibaut Weise, Mark Pauly. Example-based facial rigging. SIGGRAPH / ACM TOG 29(4): 32:1-32:6 (2010)

Example-Based Facial Rigging

Hao Li* ETH Zurich / EPFL
Thibaut Weise† EPFL
Mark Pauly‡ EPFL

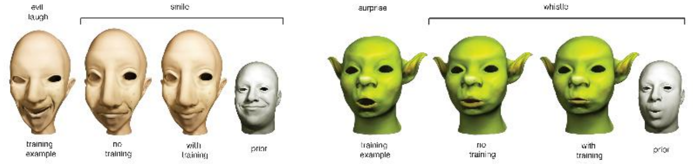


Figure 1: *Example-based facial rigging allows transferring expressions from a generic prior to create a blendshape model of a virtual character. This blendshape model can be successively fine-tuned toward the specific geometry and motion characteristics of the character by providing more training data in the form of additional expression poses.*

Abstract

We introduce a method for generating facial blendshape rigs from a set of example poses of a CG character. Our system transfers controller semantics and expression dynamics from a generic template to the target blendshape model, while solving for an optimal reproduction of the training poses. This enables a scalable design process, where the user can iteratively add more training poses to refine the blendshape expression space. However, plausible animations can be obtained even with a single training pose. We show how formulating the optimization in gradient space yields superior results as compared to a direct optimization on blendshape vertices. We provide examples for both hand-crafted characters and 3D scans of a real actor and demonstrate the performance of our system in the context of markerless art-directable facial tracking.

Keywords: facial animation, rigging, blendshape animation

1 Introduction

Building a facial model parameterization or *rig* is an essential element for animating CG characters in feature films and computer games. Due to their intuitive controls, facial rigs based on blendshape models are particularly popular among artists for creating realistic looking facial animations. Since they are art-directable, blendshape parameterizations are often used for retargeting detailed recordings of facial performances to digital faces that differ strongly from the source model. For instance, an artist has maximum control over the appearance of wrinkles and folds for a particular facial pose, as opposed to pure dynamic muscle rigs. However, hundreds of separately sculpted shapes are typically needed to achieve realism. The ability to both efficiently generate a complete customized facial rig and automatically adjust blendshapes to match the specific look of the actor's expressions (while retaining the controller semantics) is thus an important asset for the artist.

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†E-mail:thibaut.weise@epfl.ch
‡E-mail:mark.pauly@epfl.ch

This paper introduces a framework that automatically creates optimal blendshapes from a set of example poses of a digital face model. A predefined blendshape rig of a generic face is used as a prior to determine the semantics of each blendshape expression that we solve for. While in a traditional setting a precise pose needs to be provided for every blendshape, we only require a reduced set of example poses with a rough initial guess on the blending weights.

We propose an interleaved optimization that refines the blending weights and solves for the optimal blendshapes in two alternating steps. We regularize the optimization with meaningful blendshape expressions transferred from a generic face to accurately capture both the example poses and the semantics of the individual blendshapes. Expression transfer is achieved by mapping the deformation gradients between the neutral and blendshape pose triangles of the generic face to the target mesh triangles. Since we optimize for all blendshapes simultaneously, weighting local variations between the blendshapes in neutral and deformed pose is crucial in the regularization to prevent semantically incorrect blendshapes. We introduce an optimization that operates directly in gradient space in order to efficiently solve for blendshapes with semantics that correspond to those of a generic facial rig prior.

We illustrate the versatility of our system with applications to art-directable rigging for sculpted virtual characters and automated rigging from 3D scans of a real actor. A key aspect of our approach is that the blendshape reconstruction can be edited and adapted iteratively by either generating additional training expressions or adapting the blending weights of the example poses. This provides full control over the resulting blendshape model and facilitates easy integration into existing workflows such as facial tracking. Without our technique, an artist would have to adapt each blendshape to match all desired input examples.

2 Related Work

A large variety of different methods for facial rigging have been proposed in the past. Some are based on skeletons and joints

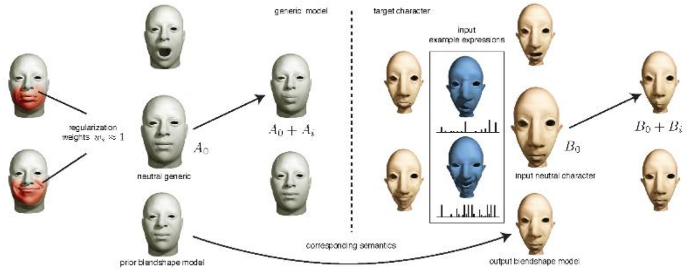


Figure 2: *Conceptual overview of our method. The generic template, illustrated with a subset of the blendshapes (left), serves as a geometry and motion prior for an actor-specific blendshape model (right). The optimization solves for the target blendshapes such that a set of example expressions are best reproduced while maintaining the semantic correspondence between template and target models. We ensure semantically correct transfer of expressions using additional per vertex regularization weights in our optimization (shown in red).*

[Magnenat-Thalmann et al. 1988], physically-based muscle models [Waters 1987], linear blendshapes [Bergeron and Lachapelle 1985], or combinations thereof. Skeleton-based rigs are most often employed for full-body animation due to their intuitive control for articulated motion. While skeletons are often used for face animation of cartoon characters, this approach is less suited to produce detailed facial expressions that exhibit wrinkles and folds. Automatic rigging using skeleton-embedding was proposed by [Baran and Popović 2007], but with the focus on full-body animation.

Physically-based muscle models are well suited for creating realistic expression dynamics and secondary motions [Sifakis et al. 2005]. However, artistic control can be difficult to achieve. [Terzopoulos and Waters 1990] proposed a semi-automatic rigging of a muscle-based model to image data. Similarly, [Kähler et al. 2001] fit a complex anatomical model to partial 3D scan data. [Orvalho et al. 2008] introduced a general rigging method by transferring a generic facial rig to 3D input scans or hand-crafted models.

Linear blendshape models [Bergeron and Lachapelle 1985] provide a good compromise between realism and control. However, hundreds of blendshapes are usually necessary to capture realistic facial expression and are often used to mimic the effect of facial muscle groups as described by Ekman's *Facial Action Coding System* [1978]. In particular, *FACS* decomposes facial behavior into 46 basic poses which are often complemented with a multitude of combined expressions and visemes. Building such a linear facial rig for highly realistic animation was recently demonstrated by [Alexander et al. 2009], though each blendshape was still hand-crafted by animators. [Pighin et al. 1998] build a rig automatically from photographs, and, similarly, [Zhang et al. 2004; Weise et al. 2009] from 3D scan data where all facial expressions are required as input. Automatic creation of facial models using (multi-) linear PCA models was proposed in [Blanz and Vetter 1999; Blanz et al. 2003; Vlasic et al. 2005], though the resulting linear blendshapes are not necessarily meaningful for facial animation control. To circumvent this problem, we propose to use a predefined generic blendshape rig as a semantic prior. This has the benefit that only a subset of expressions is sufficient to build a complete model, and moreover, the resulting blendshapes match the controller semantics of the prior.

Linear blendshape models are especially suited for retargeting [Chuang 2004]. An overview of current methods is given by [Pighin and Lewis 2006]. Choe and Ko [2005] proposed optimizing a generic predefined blendshape rig to fit sparse motion capture data of an actor. Liu et al. [2008] extended this method using expression cloning [Noh and Neumann 2001] as a prior to handle under-constrained cases where less training data is available than the number of blendshapes. In this work, we focus on building a blendshape model that has the same semantics as the input model. This is achieved by formulating the optimization problem in deformation gradient space [Sumner and Popović 2004].

3 Optimization

Our goal is to produce a full set of blendshapes from a user-provided handcrafted character or scanned 3D model in neutral expression. The user can specify an arbitrary number of additional expressions to refine the model toward the specific geometry and motion characteristics of the actor. The algorithm then determines the optimal blendshapes that best reproduce the input examples, while preserving the controller semantics and the deformation gradients of a generic blendshape model rig (c.f. Figure2). For complex input expressions, such as an angry face, it can be difficult to determine the blending weights for a given example pose and those values can vary substantially for different characters. Thus, in addition to computing the optimal blendshapes, we also solve for blending weights given a rough initial guess provided by the user.

We assume a generic blendshape model is given as a set of meshes $A = \{A_0, \ldots, A_n\}$, where A_0 is the rest pose and the A_i, $i > 0$ are additive displacements. Expressions can be generated as $T_j = A_0 + \sum_{i=1}^{n} \alpha_{ij} A_i$, where α_{ij} are the blending weights of pose T_j.

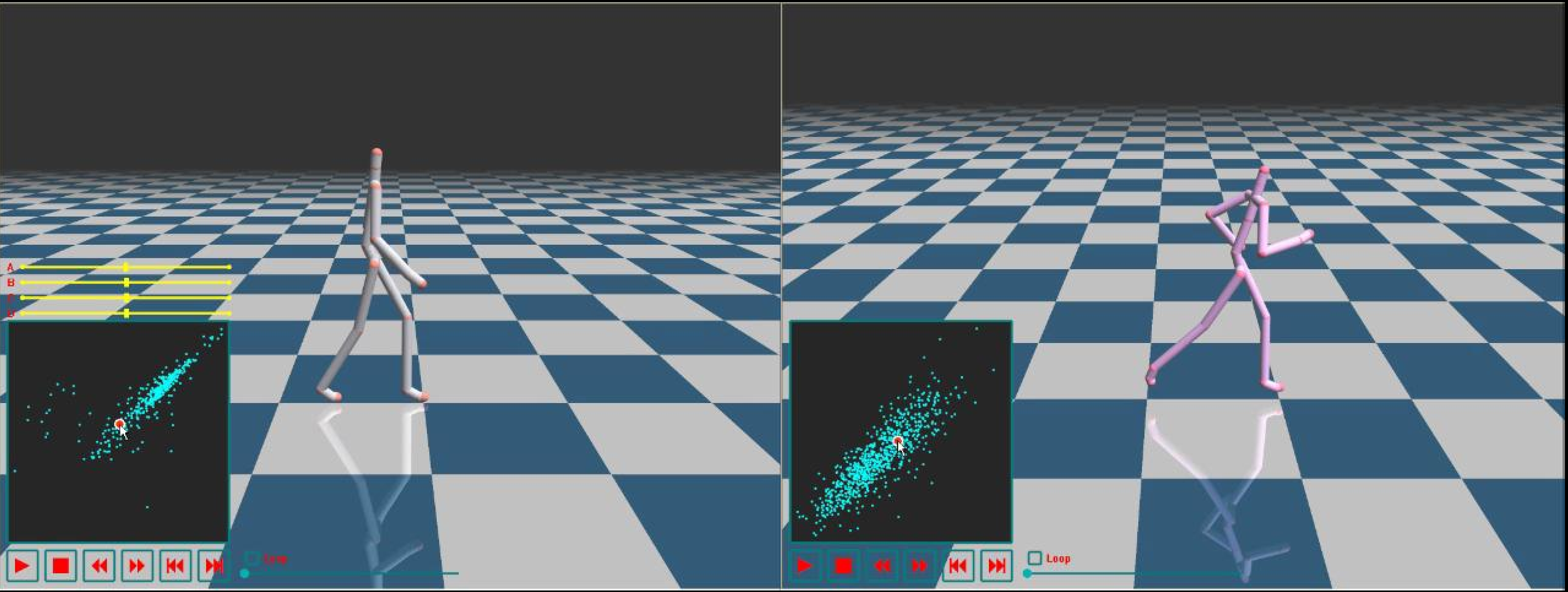
Our method is general in the sense that we can process input from various sources. In the case of 3D scans, we align the generic rest pose A_0 to the input shapes using the non-rigid registration method of [Li et al. 2009]. This produces a set $S = \{S_1, \ldots, S_m\}$ of complete meshes with connectivity of the prior model and shape of the respective scan. For hand-crafted models, we either perform

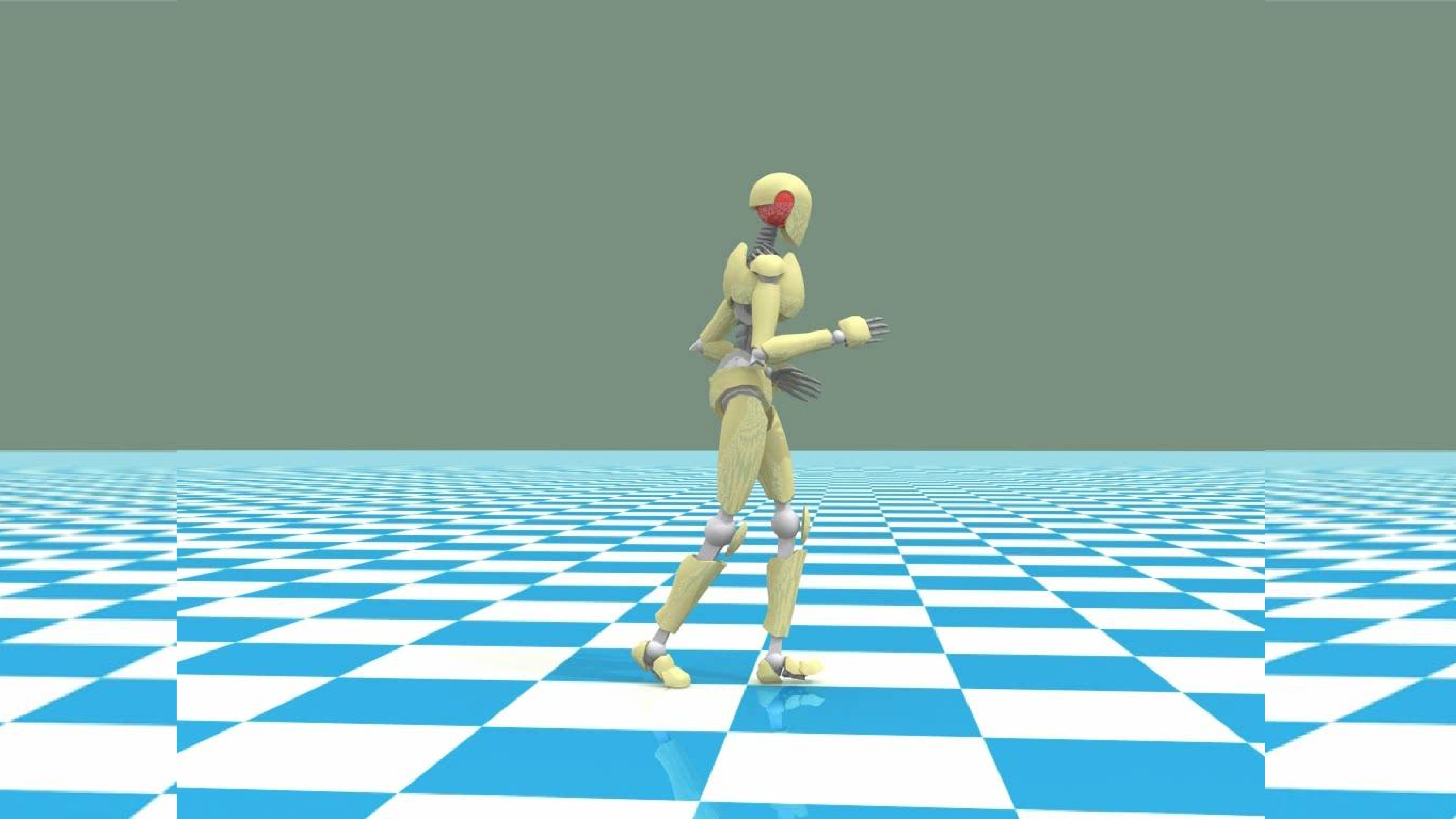
bbc.com/news/technology-34920548

Faceshift: Apple buys Star Wars motion-capture company

By Chris Baraniuk
Technology reporter

25 November 2015


A
B
Loop



Realtime Style Transfer

Online Local MAR Model

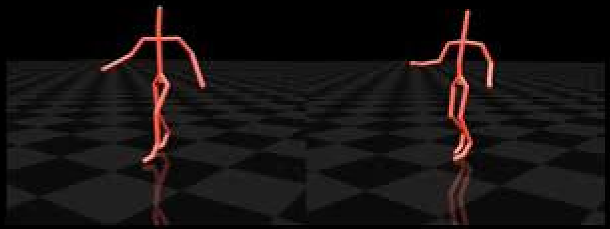
Take **unlabeled heterogeneous** motion in one style change to another style in realtime

Left: Motion capture data from internet

Right: Output animation with a new style

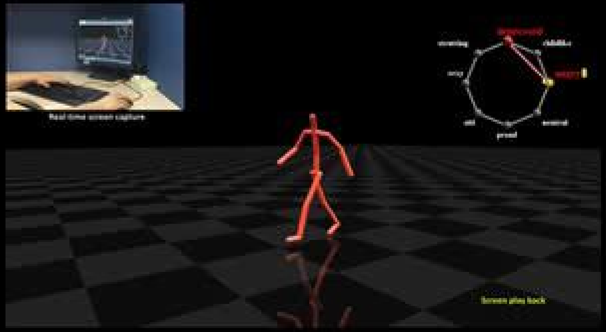
Left: Synthesized motion

Right: Output animation with a new style



Left: Keyframed motion

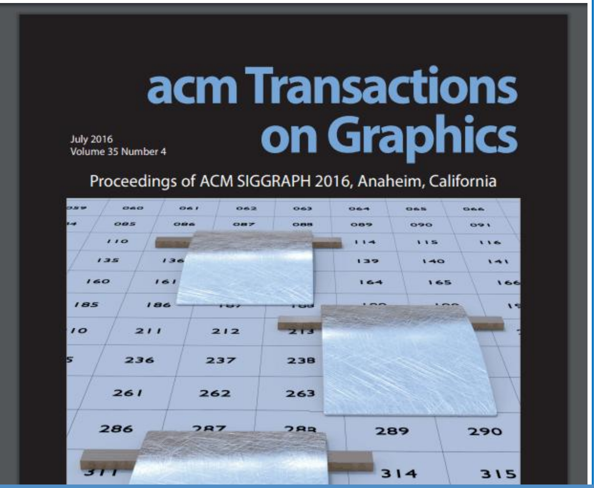
Right: Output animation with a new style

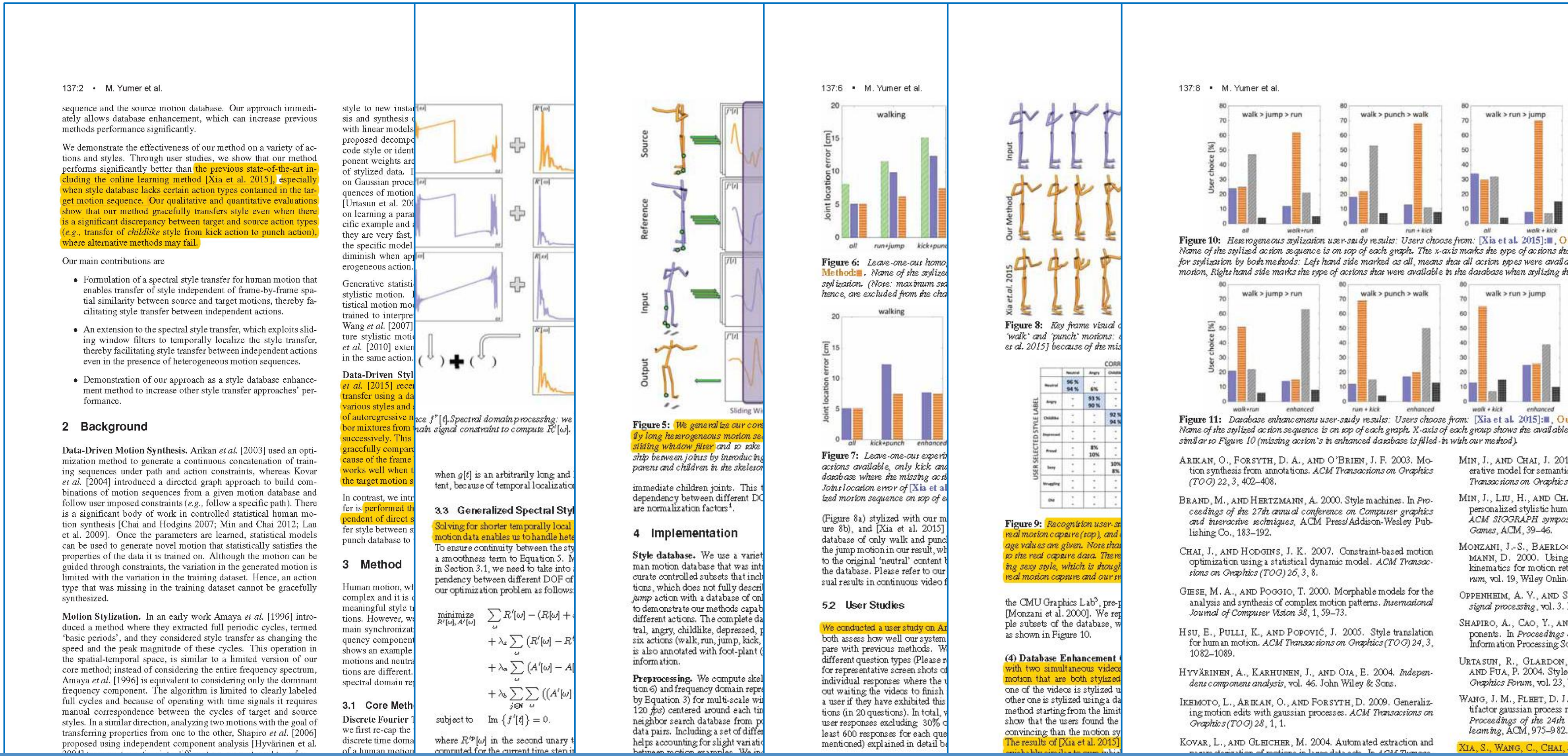


Style Control in Realtime

代表性引用与评价

ACM TOG前副编审
Niloy J. Mitra





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sequence and the source motion database. Our approach immediately allows database enhancement, which can increase previous methods performance significantly.

We demonstrate the effectiveness of our method on a variety of actions and styles. Through user studies, we show that our method performs significantly better than the previous state-of-the-art including the online learning method [Xia et al. 2015], especially when style database lacks certain action types contained in the target motion sequence. Our qualitative and quantitative evaluations show that our method gracefully transfers style even when there is a significant discrepancy between target and source action types (*e.g.*, transfer of *childlike* style from kick action to punch action), where alternative methods may fail.

Our main contributions are

- Formulation of a spectral style transfer for human motion that enables transfer of style independent of frame-by-frame spatial similarity between source and target motions, thereby facilitating style transfer between independent actions.
- An extension to the spectral style transfer, which exploits sliding window filters to temporally localize the style transfer, thereby facilitating style transfer between independent actions even in the presence of heterogeneous motion sequences.
- Demonstration of our approach as a style database enhancement method to increase other style transfer approaches' performance.

2 Background

Data-Driven Motion Synthesis. Arikan *et al.* [2003] used an optimization method to generate a continuous concatenation of training sequences under path and action constraints, whereas Kovar *et al.* [2004] introduced a directed graph approach to build combinations of motion sequences from a given motion database and follow user imposed constraints (*e.g.*, follow a specific path). There is a significant body of work in controlled statistical human motion synthesis [Chai and Hodgins 2007; Min and Chai 2012; Lau et al. 2009]. Once the parameters are learned, statistical models can be used to generate novel motion that statistically satisfies the properties of the data it is trained on. Although the motion can be guided through constraints, the variation in the generated motion is limited with the variation in the training dataset. Hence, an action type that was missing in the training dataset cannot be gracefully synthesized.

Motion Stylization. In an early work Amaya *et al.* [1996] introduced a method where they extracted full periodic cycles, termed 'basic periods', and they considered style transfer as changing the speed and the peak magnitude of these cycles. This operation in the spatial-temporal space, is similar to a limited version of our core method; instead of considering the entire frequency spectrum, Amaya *et al.* [1996] is equivalent to considering only the dominant frequency component. The algorithm is limited to clearly labeled full cycles and because of operating with time signals it requires manual correspondence between the cycles of target and source styles. In a similar direction, analyzing two motions with the goal of transferring properties from one to the other, Shapiro *et al.* [2006] proposed using independent component analysis [Hyvärinen et al.

在其SIGGRAPH论文多处、大段落地评述该工作，几乎是全文将该工作

作为基础进行了深入的比较分析和研究，并评价该工作为“state-of-the-art”。

代表性引用与评价

US010896534B1

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Smith et al.

(10) **Patent No.: US 10,896,534 B1**
(45) **Date of Patent: Jan. 19, 2021**

(54) **AVATAR STYLE TRANSFORMATION USING NEURAL NETWORKS**

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(73) Assignee: **Snap Inc.**, Santa Monica, CA (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 5 days.

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(22) Filed: **Sep. 19, 2018**

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G06T 13/40 (2011.01)
H04L 12/58 (2006.01)
G06N 3/08 (2006.01)
G06N 20/00 (2019.01)

(52) **U.S. Cl.**
CPC ***G06T 13/40*** (2013.01); ***G06N 3/08*** (2013.01); ***G06N 20/00*** (2019.01); ***H04L 51/10*** (2013.01)

(58) **Field of Classification Search**
CPC G06T 13/40; G06N 20/00; G06N 3/08; H04L 51/10
See application file for complete search history.

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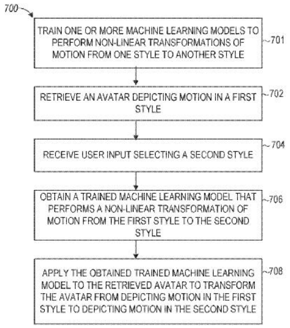
Assistant Examiner — Vincent Peren
(74) *Attorney, Agent, or Firm* — Schwegman Lundberg & Woessner, P.A.

(57) **ABSTRACT**

Aspects of the present disclosure involve a system comprising a computer-readable storage medium storing at least one program and a method for transforming a motion style of an avatar from a first style to a second style. The program and method include: retrieving, by a processor from a storage device, an avatar depicting motion in a first style; receiving user input selecting a second style; obtaining, based on the user input, a trained machine learning model that performs a non-linear transformation of motion from the first style to the second style; and applying the obtained trained machine learning model to the retrieved avatar to transform the avatar from depicting motion in the first style to depicting motion in the second style.

19 Claims, 11 Drawing Sheets



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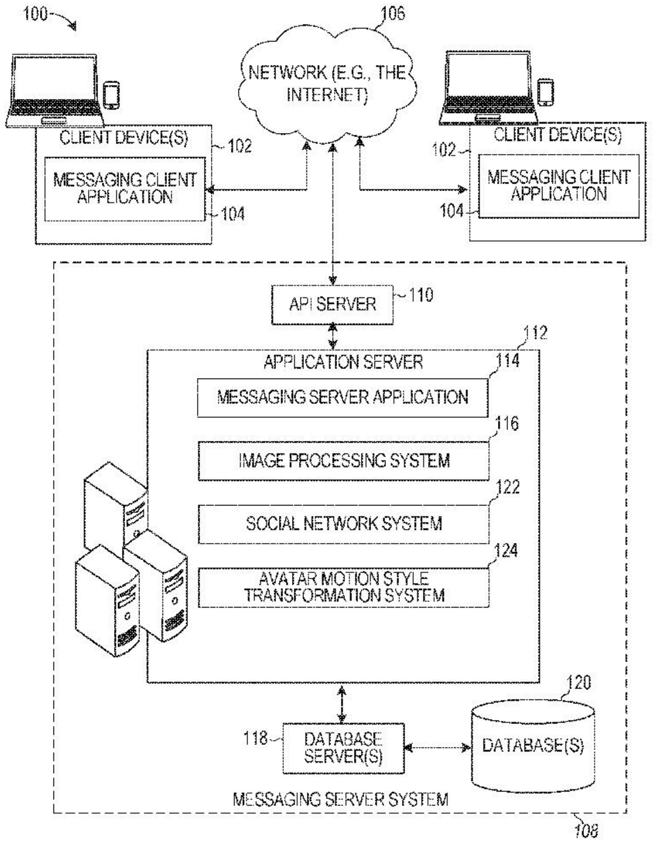

FIG. 1

谢谢！欢迎交流！